GPU substructure for flashjet

kt / C-A jet substructure on the merge history

exclusive jets · soft-drop grooming · Lund coordinates

Chirayu Gupta

Code: github.com/…/flashjet · branch `benchmarking`, commit `2e912ef`

Reading guide — features: **F1** exclusive-kt · **F2** soft-drop grooming · **F3** Lund · inputs: **A** toy events · **B** toy shower · **C** real CMS (C1 QCD, C2 ttbar-2ℓ, C3 ttbar-4q) · tree tags **kT** **C/A** **anti-kT**

Summary

Three substructure features, all **pure-torch post-reads of the existing merge history** — no kernel changes, CPU/CUDA identical, negligible cost:

	feature	reads	implements
F1	exclusive-kt jets	kT history	kt algorithm [1]
F2	soft-drop / mMDT grooming	C/A history	Soft Drop [4]
F3	Lund coordinates	C/A history	Lund plane [6]

Validated three ways, each closed against an independent reference:

validation	reference	headline
unit tests	independent NumPy tree-walks	85 passed (13 CUDA-only skipped)
paper closures (toys)	analytic LL predictions	z_g on the $1/z$ curve; areas; β-ordering
real CMS (raw-to-raw)	stored FastJet branches	p_T **1.000000**, m_{SD} **−0.004 GeV**, R_g 99.2% <0.01

⇒ **flashjet reproduces CMS's FastJet reconstruction jet-by-jet, with residuals at the level of NanoAOD float storage** — then the same histories yield **variables that separate the samples**.

What was already there vs. what we added

Pre-existing

- `history.py`: `jet_idx_from_history`, `_resolve_parents`, `_decode`, `splitting_scales_from_history`, Triton `_decode_triton`
- `reference.py`: single-event NumPy ground truth (`cluster_event`)
- `nn_reference.py`: nearest-neighbour reference
- `conftest.py`: `random_event` fixtures
- The kernels + the merge-history recording

Added (this work) — in `history.py`

- **F1** `exclusive_jets_from_history`
- **F2** `groom_from_history`
- **F3** `lund_coordinates_from_history`
- helpers: `_resolve_roots`, `_jet_roots`, `_pseudojet_p4`, `_dense_number_roots`
- `api.py`: `ClusterOutput.{exclusive_jets, groomed_jets, mass_drop, lund_coordinates}` + a `mask` field
- `tests/test_substructure.py` (new)

The `_ref_*` walkers inside `test_substructure.py` are also ours — independent naive NumPy tree-walks used only to pin the fast implementations.

The three features

	Feature	Function	Implements
F1	Exclusive jets (kt)	`exclusive_jets_from_history(...)`	kt algorithm — FastJet manual [1], anti-kt [2]
F2	Grooming (C/A)	`groom_from_history(...)`	Soft Drop [4] · mMDT [5] · mass-drop [3]
F3	Lund coordinates	`lund_coordinates_from_history(...)`	Primary Lund plane [6]

F1 — undo the last merges of the sequence to expose exactly `n_jets` (or a `d_cut`) subjets; reduces to the inclusive jets at the trivial cut.
F2 — walk each jet down the harder branch, dropping the softer prong until $z > z_{\mathrm{cut}}(\Delta R/R)^{\beta}$ (soft-drop) — the $O(\log_2 n)$ declustering.
F3 — for every primary split emit $(z, \Delta R, k_t, \ln 1/\Delta R, \ln k_t, d)$; the d-channel ties **exactly** to `splitting_scales`.

How the toys were generated

(the "simulation" — none of it pre-existing; flashjet only clusters, it does not generate events)

Input A: The toy event generators (written in the plot scripts)

flashjet takes 4-vectors in and returns jets — it has **no event generator**. To exercise the features I wrote small, transparent toys *outside the repo*, in the plot scripts.

`make_plots.py` — two single-fat-jet samples at $p_T \approx 500$ GeV, $R = 0.8$:

- **QCD-like**: one hard collinear core (92% of p_T, $\sigma_{y,\phi} = 0.06$) + soft wide-angle radiation (8%, $\sigma = 0.30$). Each "spray" = n massive pions with exponential p_T fractions.
- **W-like**: two hard prongs with pair mass $m = \sqrt{z(1-z)}\, p_T\, \Delta R = 80.4$ GeV ($z \in [0.30, 0.45]$), random orientation, + 6% soft contamination.

```
def _spray(rng, n, y0, phi0, pt_total, sigma):     # n pions around (y0,phi0)
    frac = rng.exponential(1., n); frac /= frac.sum()
    pt = pt_total * frac
    y, phi = rng.normal(y0, sigma, n), rng.normal(phi0, sigma, n)
    mt = np.sqrt(M_PI**2 + pt**2)
    return np.stack([pt*np.cos(phi), pt*np.sin(phi), mt*np.sinh(y), mt*np.cosh(y)], 1)
```

1500 events per sample. Purpose: QCD vs W is a *known* truth, so any observable that fails to separate them (or lands off the analytic curve) would be a real bug.

Input B: The toy parton shower

`make_paper_plots.py` builds a **primary, fixed-coupling, leading-log** shower — emissions sampled **uniformly in the Lund triangle** with density $\bar{\alpha}$ per unit $(\ln 1/\theta,\ \ln k_t)$ area, off a hard spine at $(y, \phi) = (0, \pi)$:

$$\theta = e^{-u}, \quad k_t = e^{v}, \quad z = \frac{k_t}{p_{T0}\,\theta}, \qquad \text{keep } v \le \ln(p_{T0}/2) - u \ (z \le \tfrac{1}{2}),\ k_t > k_t^{\min}$$

- **Jet areas**: one hard event + a dense grid of infinitely-soft **ghosts** ($p_T = 10^{-8}$); the ghosts each algorithm sweeps up trace its catchment area (the anti-kt-paper construction).

$\bar{\alpha} = 0.25$, $p_{T0} = 1$ TeV, $R = 0.4$, 20 000 showers. Everything is re-runnable with a fixed seed (`20260713`).

Input C: the real CMS datasets

Real detector-level **PF-candidate constituents** — the only NanoAOD flavour (JMENano, 150X) that stores the `PFCand` table + the `FatJetPFCand` constituent→AK8 map. Pulled via DAS/`xrdcp`.

tag	dataset	era	jets	used in
C1	`QCD_Pt-15to7000_Flat2018_pythia8`	UL18 JMENano (13 TeV)	60 257	CMS (1)–(4), β-family
C2	`TTTo2L2Nu` (dileptonic)	UL18 JMENano (13 TeV)	12 561	ttbar slides
C3	`TTto4Q` (fully-hadronic)	RunIII2024 JMENanoV15 (13.6 TeV)	16 516	three-sample compare, **separating variables**

All: leading AK8 jet/event, raw $p_T > 300$ GeV, $|\eta| < 2.4$, 2–200 linked PF candidates.

`PFCand_pt/mass` are **already PUPPI-weighted** (branch titles); stored `FatJet_*` are JEC-corrected (raw = `pt×(1−rawFactor)`). AK4 uses **MINIAOD** instead (see backup) — no AK4 constituent linker exists in NanoAOD.

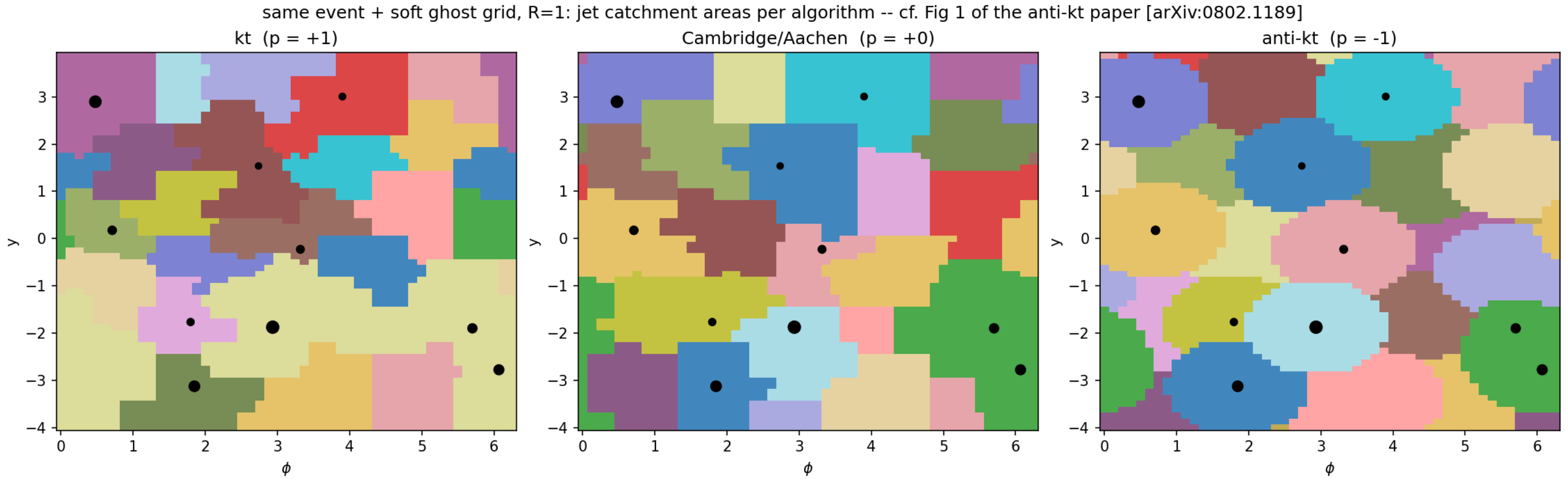
Correctness

each feature reproduces the signature figure of the paper it implements

Bottom line: on toy inputs with known truth, F1/F2/F3 land on the papers' analytic predictions.

anti-kt jet shapes — reproduces [2] Fig. 1

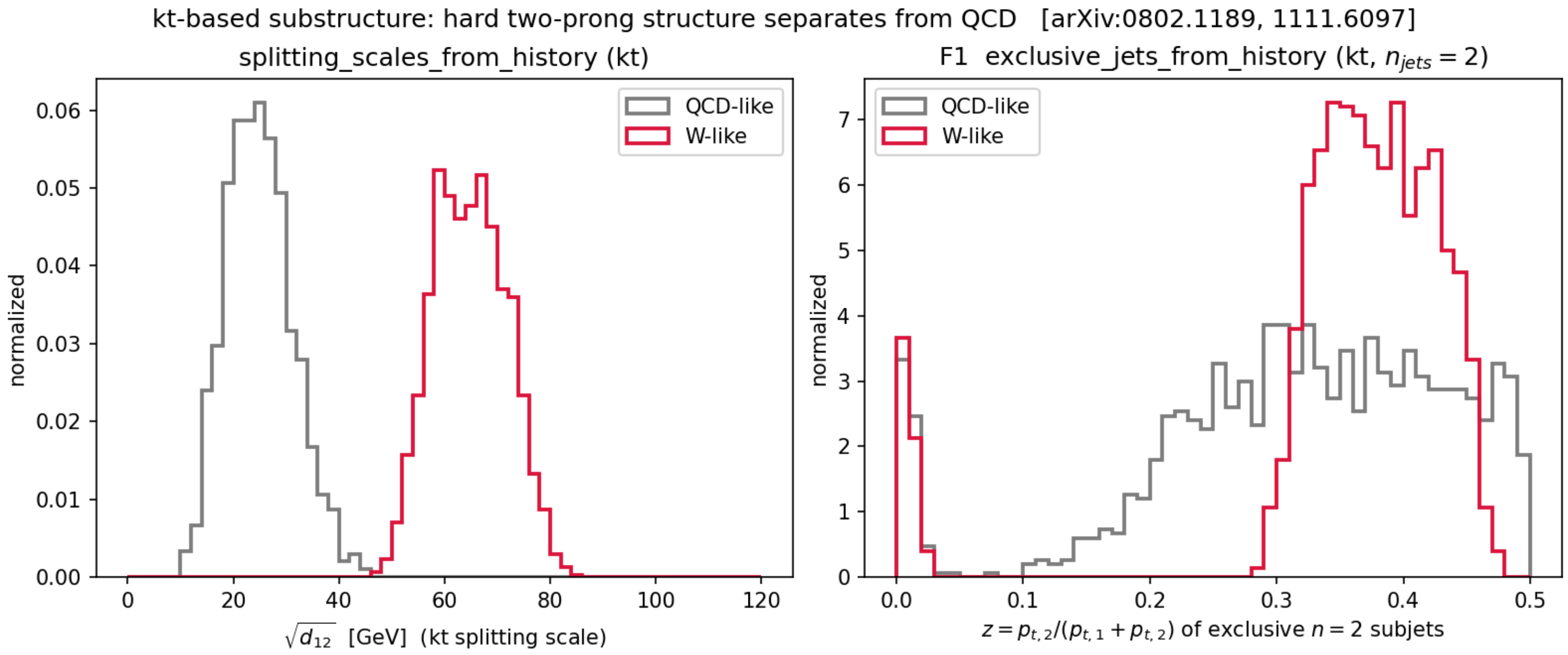
Input B (toy shower), clustering only — no substructure feature



What: each algorithm's **jet catchment area**. kt & C/A give ragged, area-fluctuating jets; **anti-kt** gives rigid **circles** around each hard particle — the defining anti-kt result, from flashjet's own clustering. 1 synthetic event (10 hard particles + ~3200 ghosts), $R = 1$.

F1 — kt substructure separates 2-prong from QCD [1, 2]

Input A (QCD/W toy fat-jets), kt clustering



Left — $\sqrt{d_{12}}$ (`splitting_scales_from_history`): kt scale of the last merge.
W-like peaks at $\sim m_W/2$, QCD low.

Right — exclusive 2-subjet z (`exclusive_jets_from_history`, $n_{\text{jets}} = 2$):
W-like balanced ($z \approx 0.35$), QCD lopsided ($z \to 0$).

F2 — soft-drop z_g vs the analytic prediction [4]

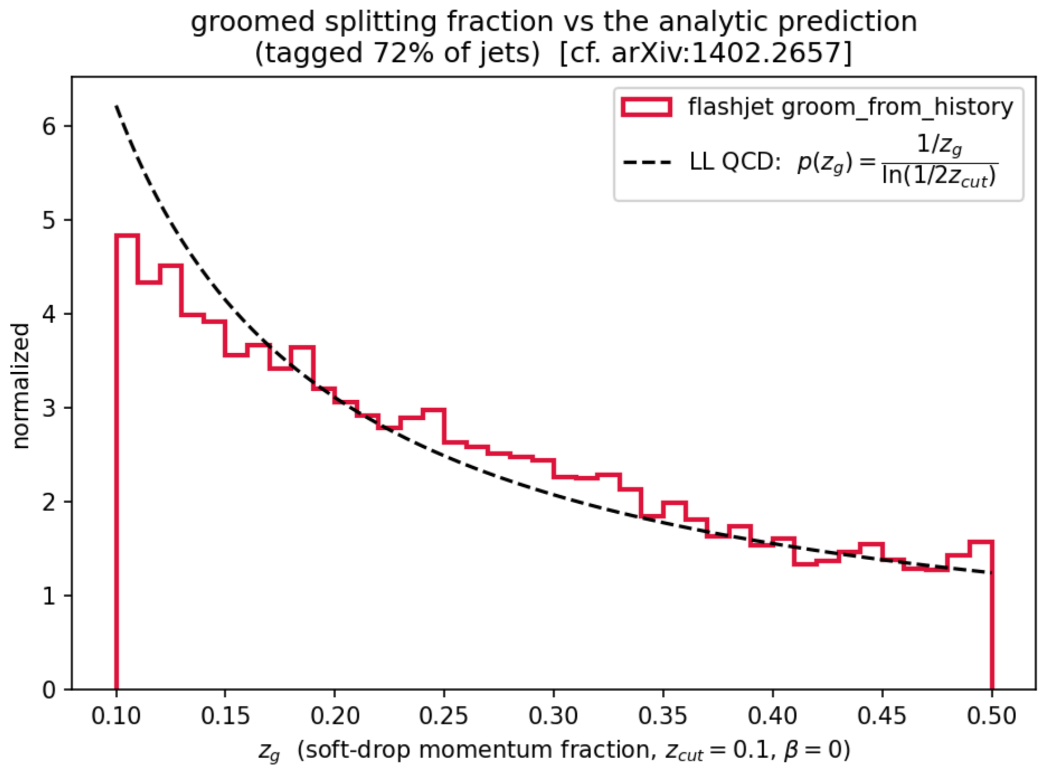
Input B (toy shower), C/A soft-drop

The soft-drop momentum fraction z_g from `groom_from_history` ($z_{\rm cut} = 0.1,\ \beta = 0$) tracks the **leading-log QCD prediction**

$$p(z_g) = \frac{1/z_g}{\ln(1/2z_{\rm cut})}$$

across the **entire range**, no free parameters. 72% of jets tagged.

> **Decisive grooming-correctness plot**: the target curve is *the* right
> answer, so lying on it is unambiguous closure.



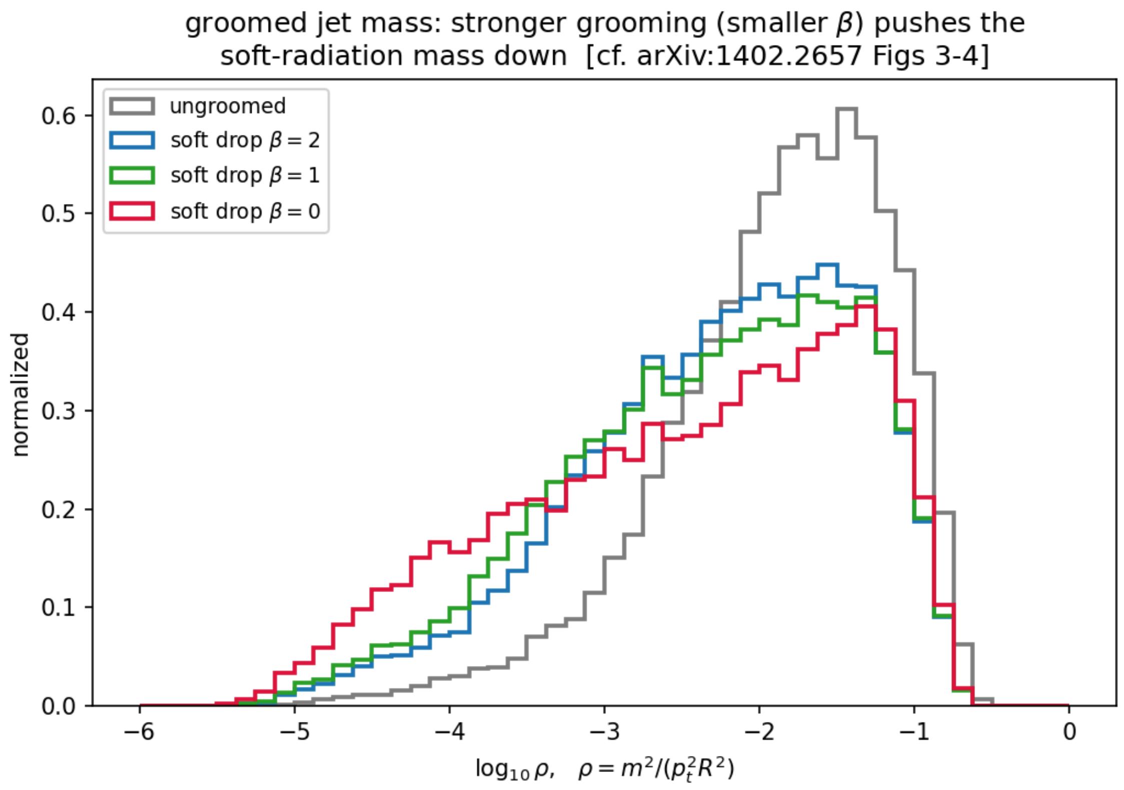
F2 — groomed mass ordering in β [4] Figs. 3–4

Input B (toy shower), C/A soft-drop at $\beta = 0, 1, 2$

Groomed mass $\rho = m^2/(p_T^2 R^2)$ for $\beta = 0, 1, 2$ vs ungroomed.

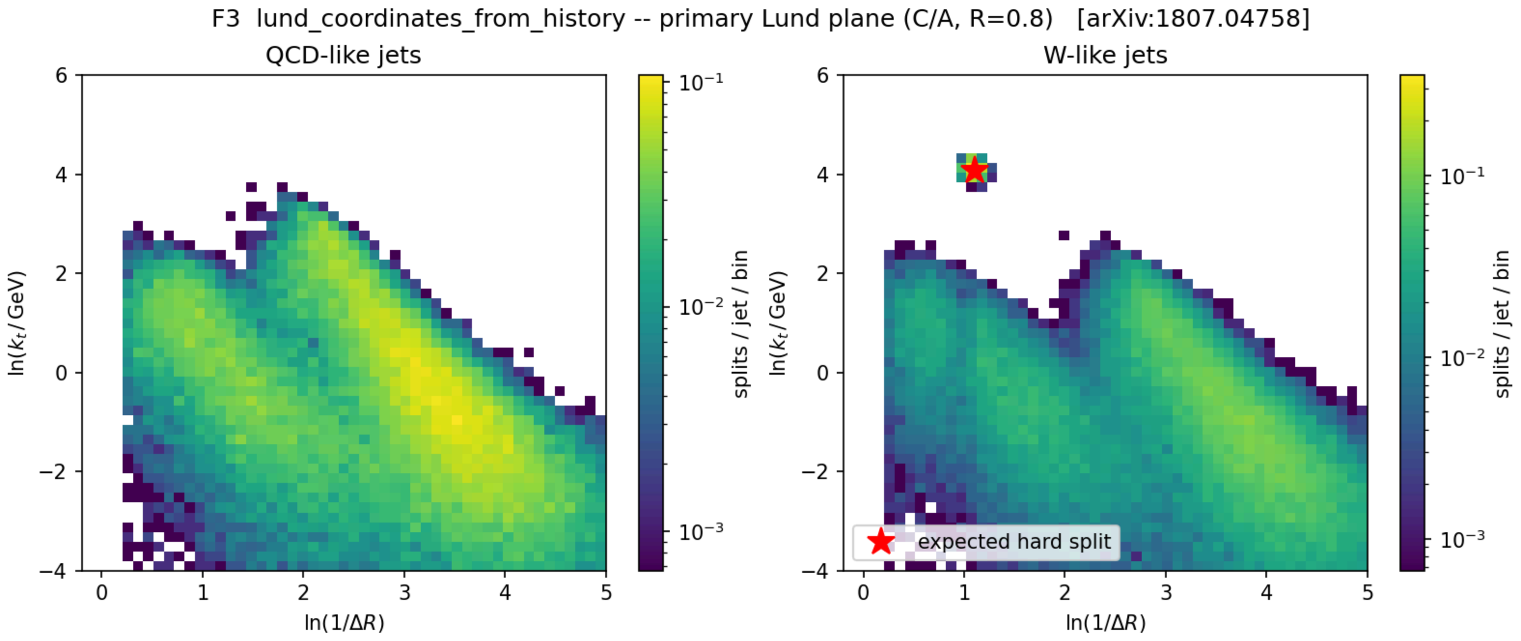
- Grooming shifts the spectrum to **lower** ρ (removes soft-wide radiation).
- **Smaller β grooms harder**: the $\beta = 0$ curve reaches furthest left.

Reproduces the β-ordering of the Soft Drop paper.



F3 — primary Lund plane [6]

Input A (QCD/W toy fat-jets), C/A clustering



`lund_coordinates_from_history` (C/A, $R = 0.8$). **QCD** fills the soft-collinear region smoothly. **W-like** shows the same background **plus an isolated hard-splitting spot** exactly where the 2-body m_W decay must sit (red ★ = predicted position). [6]

On real CMS data (Input C)

not toys any more: real detector-level PF-candidate constituents

Bottom line: on real constituents, flashjet reproduces CMS's stored FastJet output *jet-by-jet, raw-to-raw*, with the small residuals at the level of NanoAOD float storage.

The real-data pipeline (`make_cms_plots.py`) — dataset C1

These plots use *real jet constituents*, not toys:

1. Pick every CMS **AK8 FatJet** with $p_T > 300$, $|\eta| < 2.4$.
2. Gather its **PF candidates** via `FatJetPFCand_jetIdx→pfCandIdx→PFCand_{pt,η,φ,m}`. **`PFCand_pt/mass` are already PUPPI-weighted** (branch titles) — no separate weight branch.
3. Feed *those constituents* to **our** `cluster(R=0.8, antikt)` — CMS AK8 **is** anti-kt $R = 0.8$.
4. Leading jet → **F2** soft drop ($z_{\text{cut}} = 0.1, \beta = 0$, declustering the **C/A** tree, as FastJet's SoftDrop does) — and **F3** Lund.
5. Chunked at 3000 jets (torch backend is $O(N^3)$; each AK8 jet is one independent event, so chunking is exact).

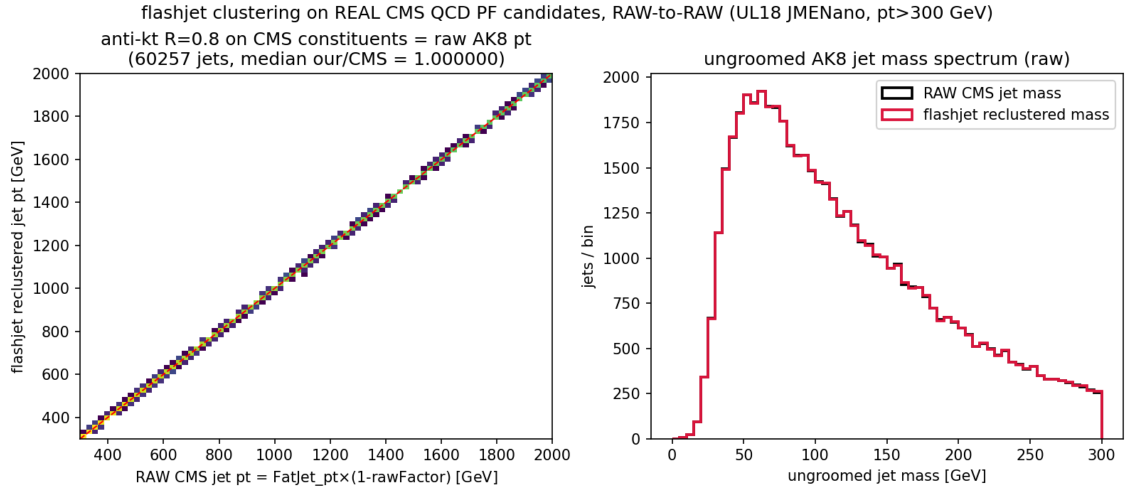
The stored CMS values are JEC-corrected: raw jet = `FatJet_pt×(1−rawFactor)`; `FatJet_msoftdrop` = m(sub1+sub2) **with subjet JECs** (checked against the data: Δ = +0.0002 GeV).

CMS (1) — reclustering closes: our p_T vs CMS `FatJet_pt` — C1

What: feed CMS's own AK8 constituents to **our** anti-kt $R = 0.8$ and compare the reclustered jet p_T to CMS's stored `FatJet_pt`, **jet-by-jet**.

Result: vs the **raw** jet pt (`FatJet_pt×(1−rawFactor)`) the ratio is **median 1.000000, σ = 2.5×10^{-4}** — the spread is consistent with NanoAOD float storage. The apparent ~6% offset vs the stored value **is the L1L2L3 JEC**: CMS stores corrected p_T, we recompute the raw one.

60 257 jets; HTCondor cluster 9087059.

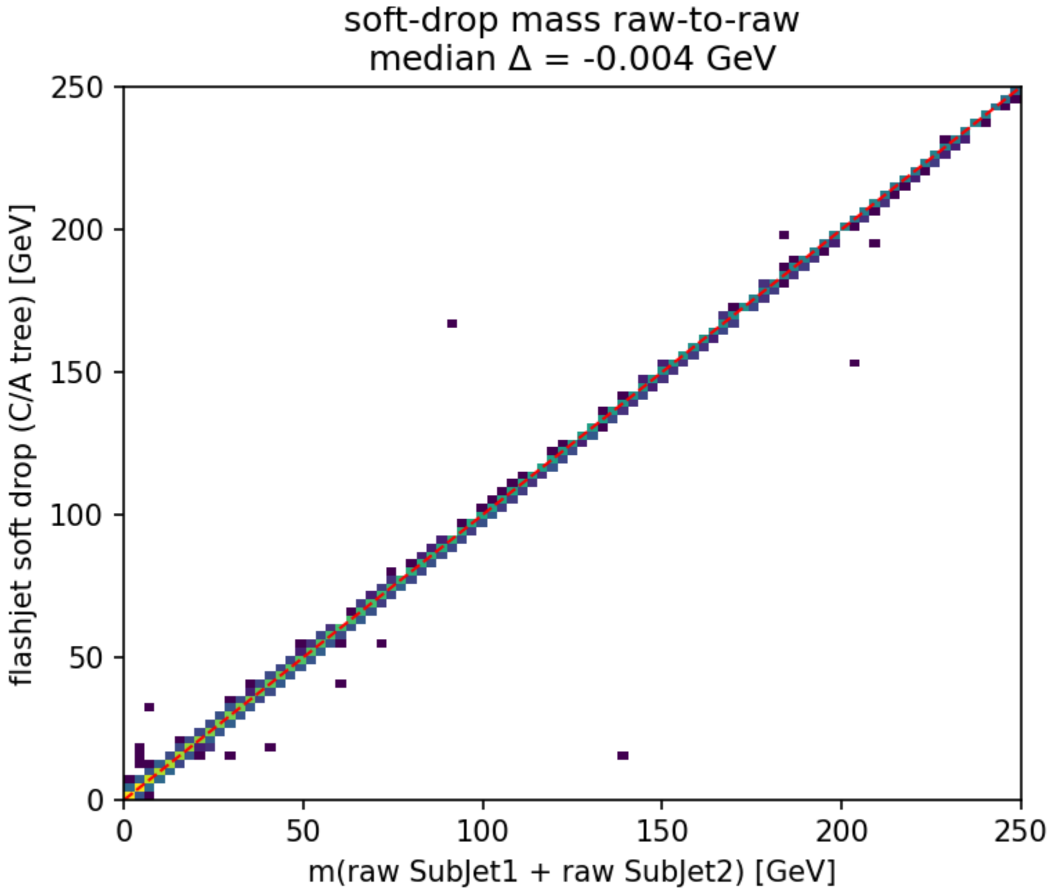


CMS (2) — soft-drop mass matches CMS `msoftdrop` (raw-to-raw) — C1

Raw-to-raw, on the C/A tree (20 065 jets): our **F2** soft-drop mass reproduces CMS `FatJet_msoftdrop` jet-by-jet — the diagonal is tight, with residuals near the NanoAOD storage level.

	our − CMS
m_{SD}	**−0.004 GeV** (95.6% <0.5 GeV)
z_g	$\lvert\Delta\rvert$ = **7×10^{-5}**

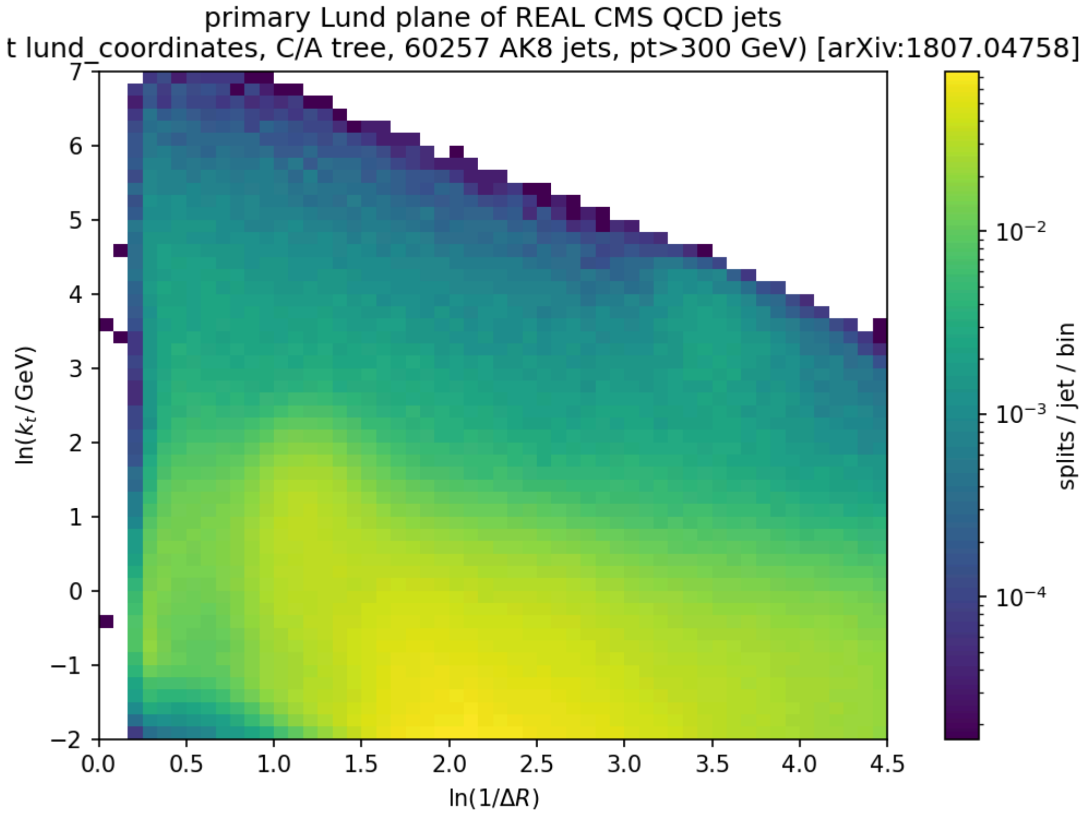
F2 reproduces CMS `msoftdrop`. The residual 4.4% tail is *consistent with* NanoAOD storage effects (see backup — soft-candidate table floor + rounding), though we don't claim that accounts for all of it.



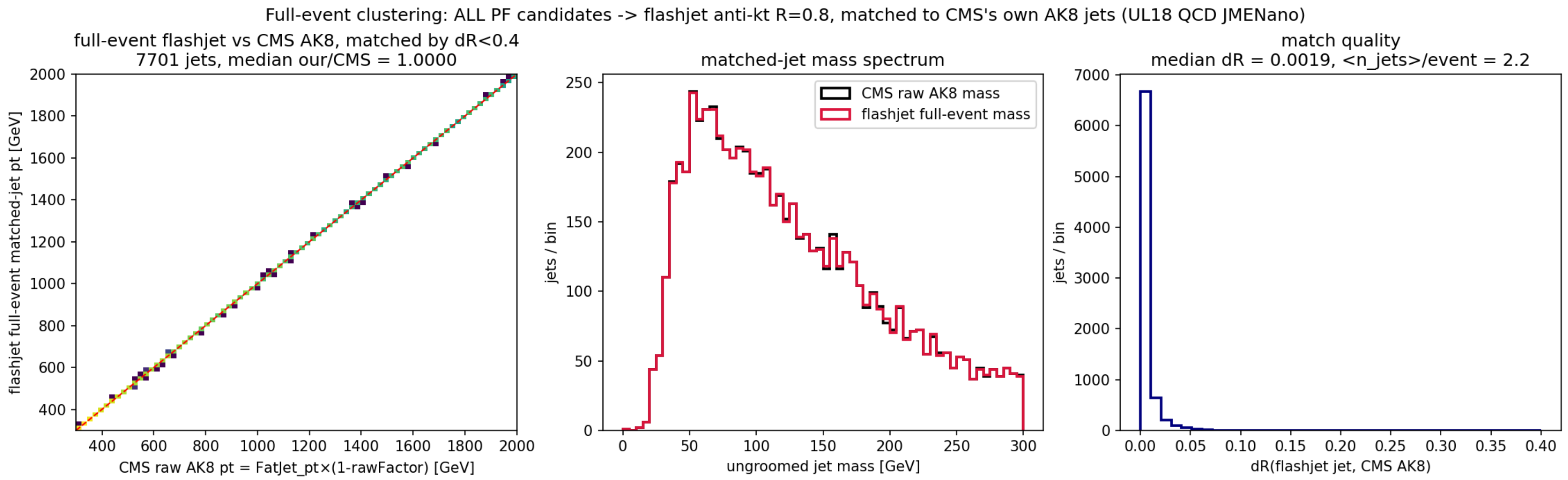
CMS (3) — primary Lund plane of 60 257 real QCD jets — C1

What: F3 `lund_coordinates` on the same real AK8 jets. Needs no comparison curve — it *is* a clean, publication-quality primary Lund plane straight from detector-level simulation.

The full [6] structure emerges with **no toy input**: the hard-collinear perturbative ridge, the soft plateau, and the three kinematic edges.



CMS (4) — full-event clustering recovers CMS's own AK8 jets — C1



What: the realistic path — feed **every PF candidate in the event** to flashjet anti-kt $R = 0.8$ (no `FatJetPFCand` per-jet pre-grouping), then match the resulting inclusive jets to CMS's **stored** AK8 jets by $\Delta R < 0.4$.

Result (7 701 matched jets, $p_T > 300$, raw-to-raw): pt **median 1.0000, 100% within 2%**; match ΔR **median 0.0019**; mass spectra agree closely. Full-event flashjet recovers CMS's own FastJet AK8 jets to milliradian ΔR — no jets known a priori.

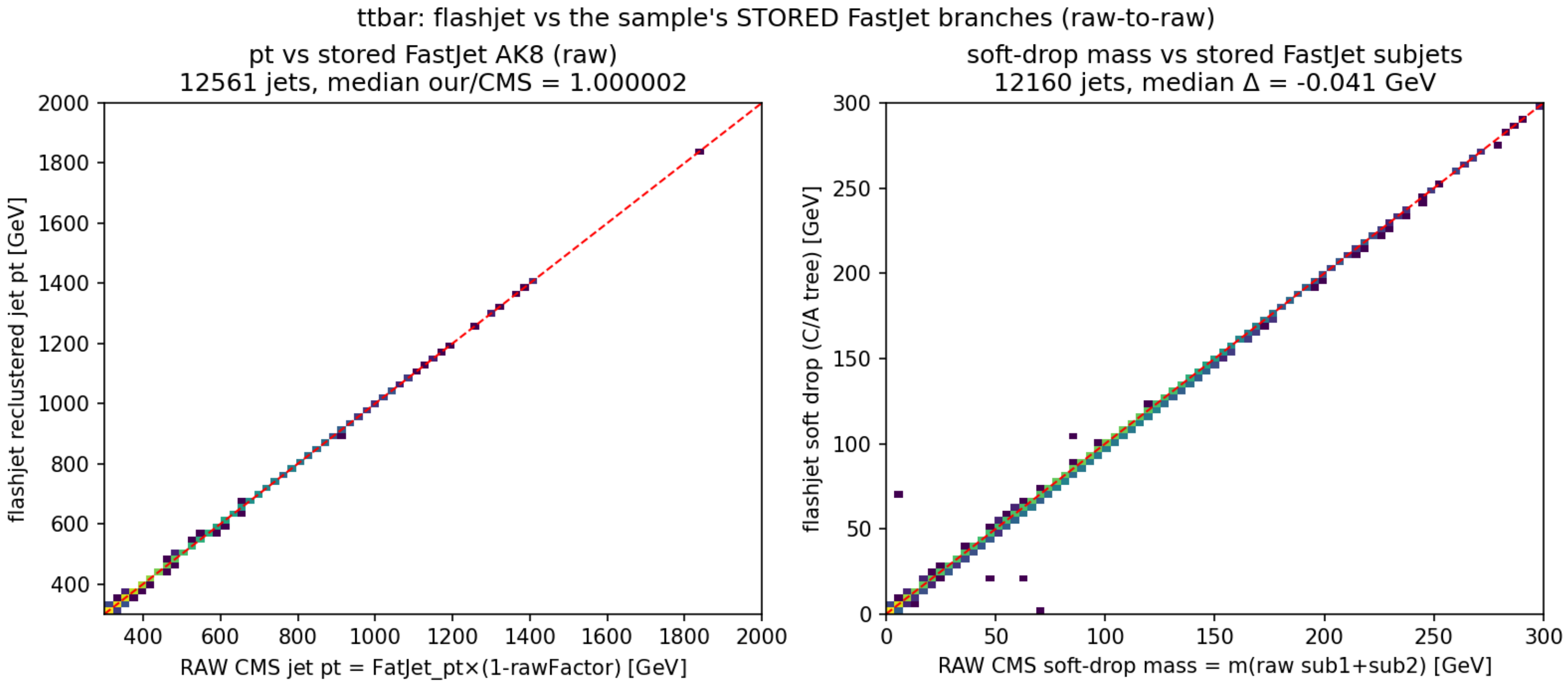
On a ttbar sample — dataset C2

a second, independent final state · comparison against the sample's own stored FastJet branches

`TTTo2L2Nu` UL18 JMENano 150X, 344 k events → 12 561 leading AK8 jets. "Comparison against FastJet" = against CMS's **stored** `FatJet_*` / `SubJet_*` / `tau*` branches, **not** a FastJet re-run.

`TTTo2L2Nu` is **dileptonic** → these jets are mostly **b-jets + ISR**, not hadronic top decays.

ttbar — flashjet ≈ stored FastJet branches (raw-to-raw) — C2

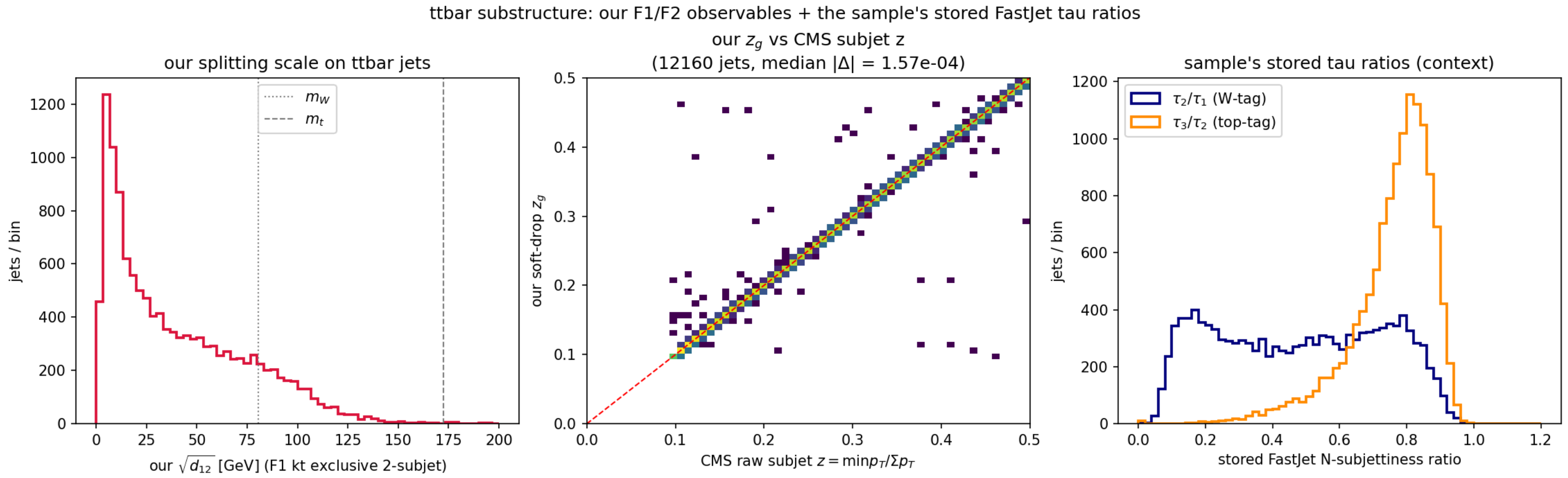


pt (our anti-kt $R = 0.8$ vs `FatJet_pt×(1−rawFactor)`): median **1.000002**, σ 2.2×10^{-4}.

soft-drop mass (our C/A tree vs m(raw sub_1+sub_2)): median Δ **−0.041 GeV**, 94.2% <0.5 GeV.

The close agreement holds on a completely different final state (boosted tops + b-jets), not just QCD — same conclusion, residuals again near the NanoAOD storage level.

ttbar — our substructure vs stored FastJet observables — C2



Left (F1): our exclusive-2-subjet $\sqrt{d_{12}}$ on ttbar jets; the m_W/m_t lines are **mass-scale references only** (dileptonic sample has no hadronic decays).
Middle (F2): our soft-drop z_g vs CMS's **raw subjet** z, jet-by-jet — median |Δ| = **1.6×10^{-4}**.

Right (context): the sample's stored FastJet **N-subjettiness** ratios — τ_2/τ_1 (W-tag), τ_3/τ_2 (top-tag). Our z_g tracking CMS's stored subjet z to 10^{-4} closes F2 directly against FastJet's SoftDrop output.

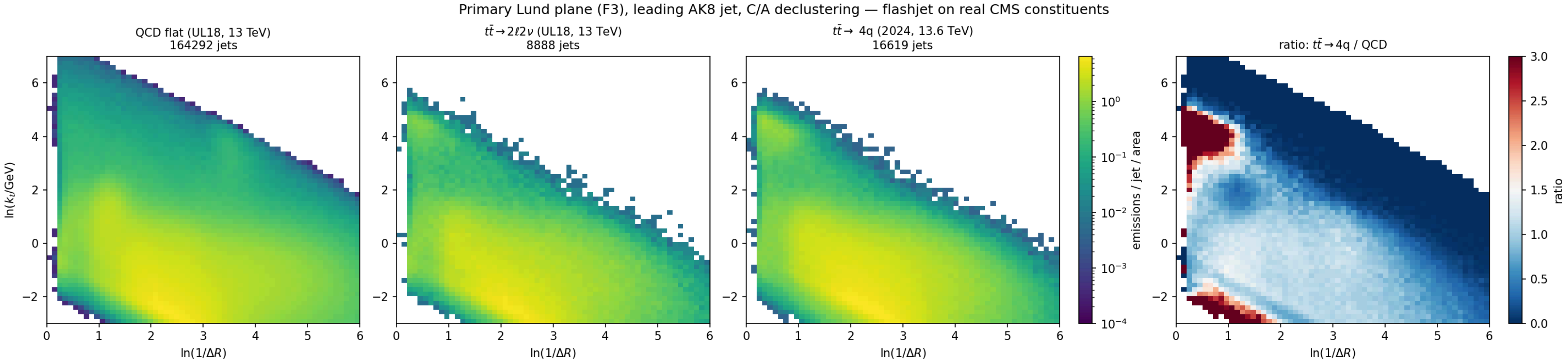
Three samples side by side

*QCD (C1) vs dileptonic ttbar (C2) vs **fully-hadronic** ttbar (C3) — the first sample with real boosted W/top decays*

C1 160 393 jets · C2 8 639 jets · C3 (`TTto4Q` RunIII2024 JMENanoV15, 13.6 TeV) 16 516 jets.

All: leading AK8 jet/event, raw $p_T > 300$, $|\eta| < 2.4$, ≤200 constituents (`make_compare_plots.py`, HTCondor 9099026).

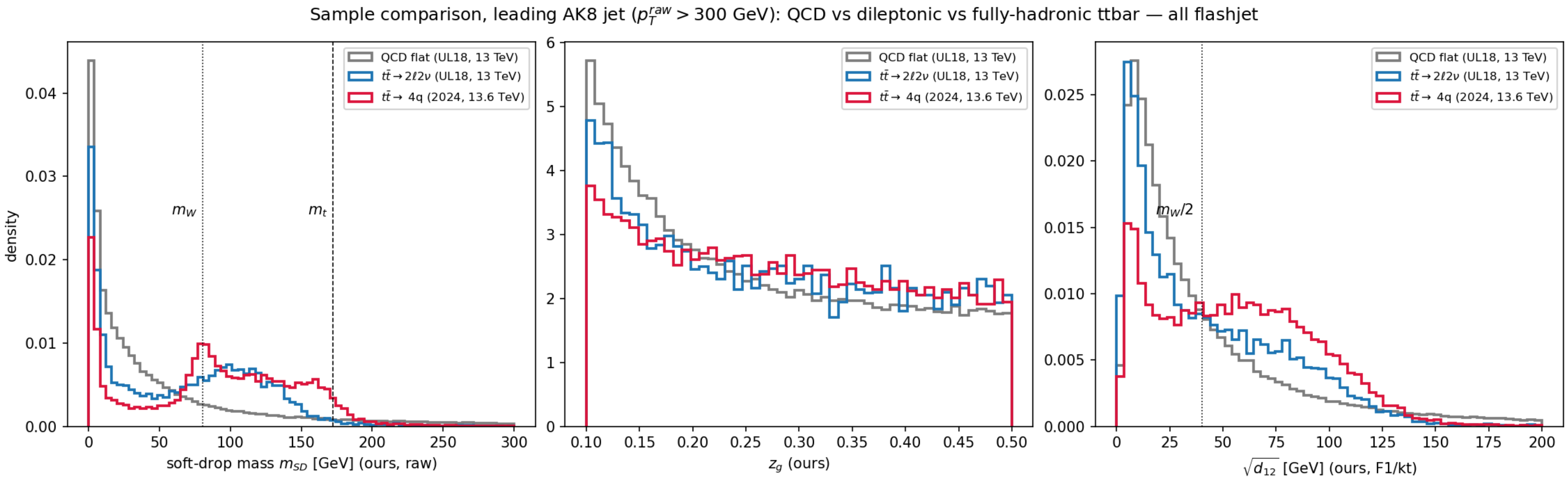
Lund planes: QCD → b-jets → boosted tops (F3) — C1/C2/C3



The top-decay scale appears exactly where it must: the $t\bar{t} \to 4q$ plane (C3) grows a hard-splitting blob at $\ln k_t \approx \ln(m_W/2) \approx 3.7$, wide-angle — over **2×** QCD in the ratio panel (right).

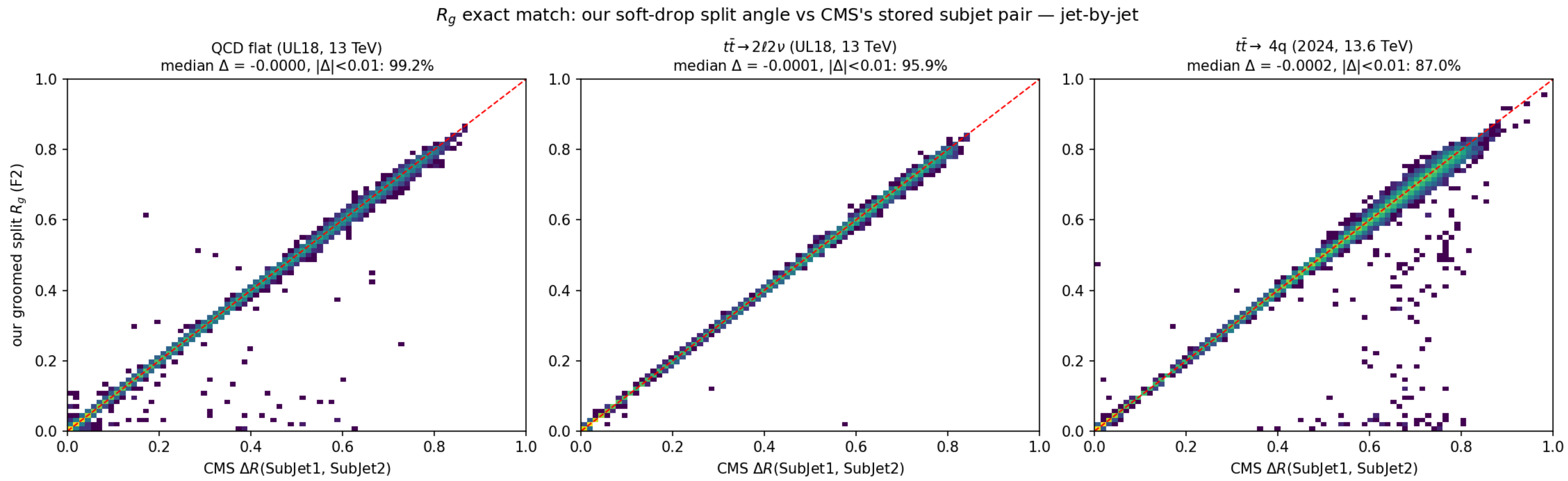
The dileptonic sample (C2, b-jets, no hadronic top) shows only a mild version. Same clustering, same F3, three physics regimes.

Spectra: m_W / m_t peaks from our grooming (F1 + F2) — C1/C2/C3



Left (F2): our raw soft-drop mass — C3 peaks at m_W with a top shoulder at m_t; C2 broad and lower; C1 falls. **Middle (F2):** z_g — ttbar flatter than QCD's $\sim 1/z$. **Right (F1):** $\sqrt{d_{12}} = m\sqrt{z/(1-z)}$ — W-window jets peak at **39** ≈ $m_W/2$, top-window at **81** ≈ $m_t/2$; QCD collapses to low scales.

R_g — a second jet-by-jet close match (F2 vs stored subjets) — C1/C2/C3



What: our split angle R_g (`groom_from_history`'s `dR`) vs stored ΔR(sub$_1$,sub$_2$), **jet-by-jet**: median Δ ≤ 2×10^{-4} — with z_g, both grooming observables close.

Run 3 (C3) = the same mechanism at higher pileup: relative agreement stays **0.14%**.

sample	p_T ratio	m_{SD} Δ	R_g |Δ|<0.01
C1 QCD UL18	0.999999	−0.004 (95.7%<0.5)	**99.2%**
C2 $t\bar{t}$ 2ℓ2ν	0.999998	−0.033 (94.1%<0.5)	95.9%
C3 $t\bar{t}$ 4q '24	0.999556	−0.077 (69.3%<0.5)	87.0%

Variables that separate the samples

the same histories that reproduce FastJet also carry physics that separates QCD from boosted decays

Bottom line: the merge history yields variables that cleanly separate boosted top/W jets from QCD — and localizes *where* they help (boosted 2-/3-prong decays, not AK4 flavour).

Which tree does each variable come from?

Each jet is clustered **three times** (`extract_tagger_vars.py`); every input is a post-read of one history:

anti-kT $R = 0.8$ — the physical jet

`algorithm="antikt"` → leading-jet 4-vector

- m_{ung} (ungroomed mass), n_{const}

kT **exclusive splitting scales**

`algorithm="kt"` → `splitting_scales()`

- $\sqrt{d_{12}}, \sqrt{d_{23}}, \sqrt{d_{34}}$ (value-sorted)
- ⇒ ratios d_{23}/d_{12}, f_{21}, f_{32}

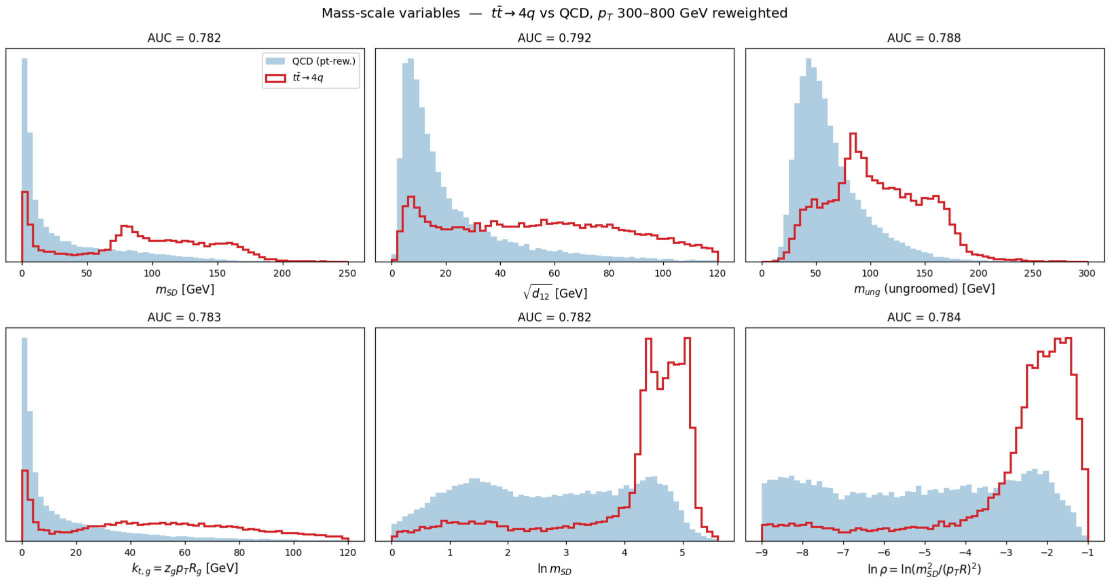
C/A **big-R recluster → soft-drop + Lund**

`algorithm="cambridge"` → `groomed_jets` + `lund_coordinates`

- groom: $m_{SD}, z_g, R_g, n_{drop}$
- Lund: $n_{Lund}, n(k_t > 1), n(k_t > 5)$, $\ln k_t^{(1,2,3)}$, z, ΔR of hardest emission

Why two trees? kT is *value-sorted* — its exclusive scales $\sqrt{d_{12}} \geq \sqrt{d_{23}} \geq \ldots$ read off the hardest splittings directly (prong hierarchy). C/A is *angular-ordered* — its primary declustering **is** the Lund plane / soft-drop sequence. The colored tags below mark each variable's source.

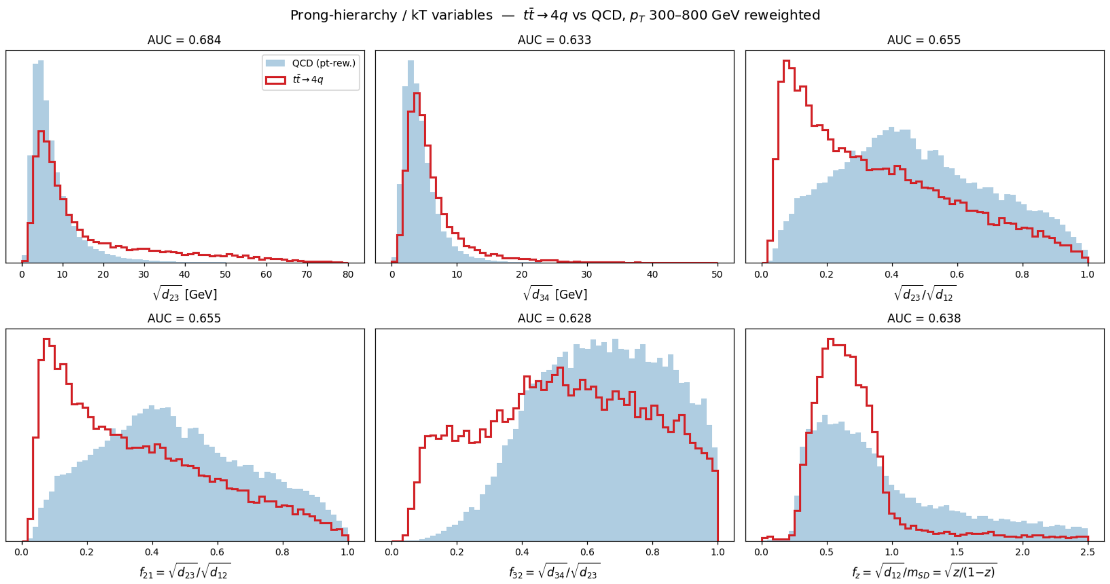
All inputs (1/4) — mass-scale variables



Filled = QCD (pt-reweighted), red = $t\bar{t} \to 4q$ (C3); per-panel weighted single-variable AUC. All at **0.78–0.79** and 0.8–0.97 correlated — they probe the *same* 2-prong decay mass, so add little to one another.

- C/A m_{SD} (0.782): soft-drop mass — clean $m_W \approx 80$ peak + $m_t \approx 160$ shoulder; QCD a steep low-mass continuum.
- kT $\sqrt{d_{12}}$ (0.792, best of group): momentum-weighted mass scale of the hardest split.
- anti-kT m_{ung} (0.788): ungroomed mass — signal keeps its mass, QCD's is inflated by soft wide radiation.
- C/A $k_{t,g} = z_g p_T R_g$ (0.783): k_t of the groomed split, another mass proxy.
- C/A $\ln m_{SD}$ (0.782), $\ln \rho$ (0.784): log forms. QCD ~flat in $\ln \rho$; signal piles at the decay mass.

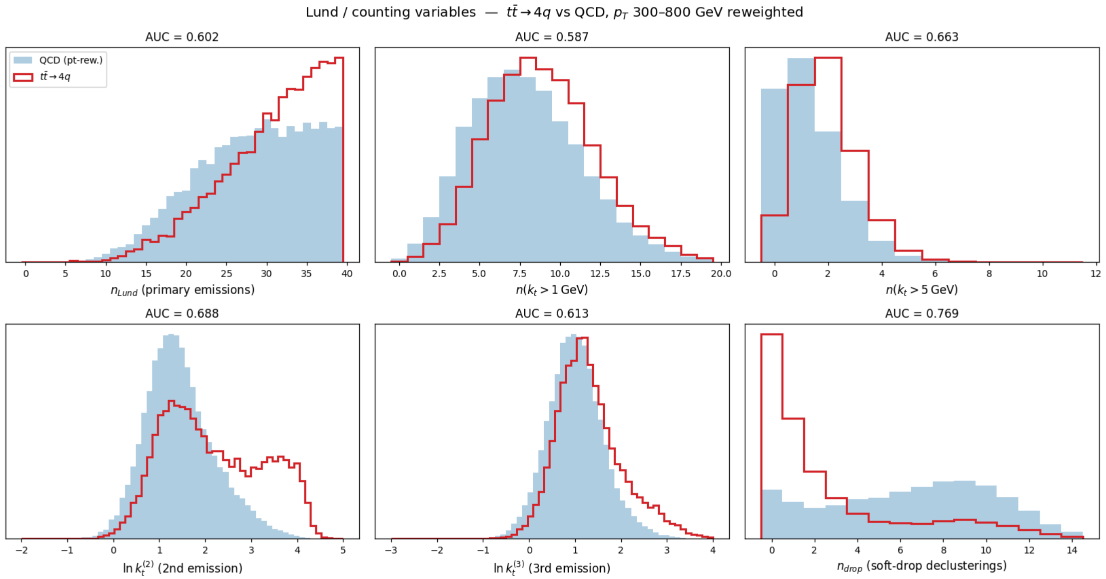
All inputs (2/4) — prong-hierarchy / kT variables



The kT splitting scales *beyond the first* and their **ratios** — do multiple hard prongs exist with a hierarchy (3-prong top, 2-prong W) vs QCD's single DGLAP ladder?

- kT $\sqrt{d_{23}}$ (0.684): 2nd kT scale — populated for 3-prong top (W→qq̄ inside), near zero for 2-prong W or 1-prong QCD.
- kT $\sqrt{d_{34}}$ (0.633): 3rd splitting — weakest, mostly extra radiation.
- kT d_{23}/d_{12} = f_{21} (0.655): the ratio removes overall scale — signal peaks near 0.1, QCD broad.
- kT $f_{32} = \sqrt{d_{34}}/\sqrt{d_{23}}$ (0.628): next ratio in the hierarchy.
- kT÷C/A $f_z = \sqrt{d_{12}}/m_{SD} = \sqrt{z/(1-z)}$ (0.638): momentum sharing, **mass-decorrelated** — decay shares evenly, QCD soft-biased.

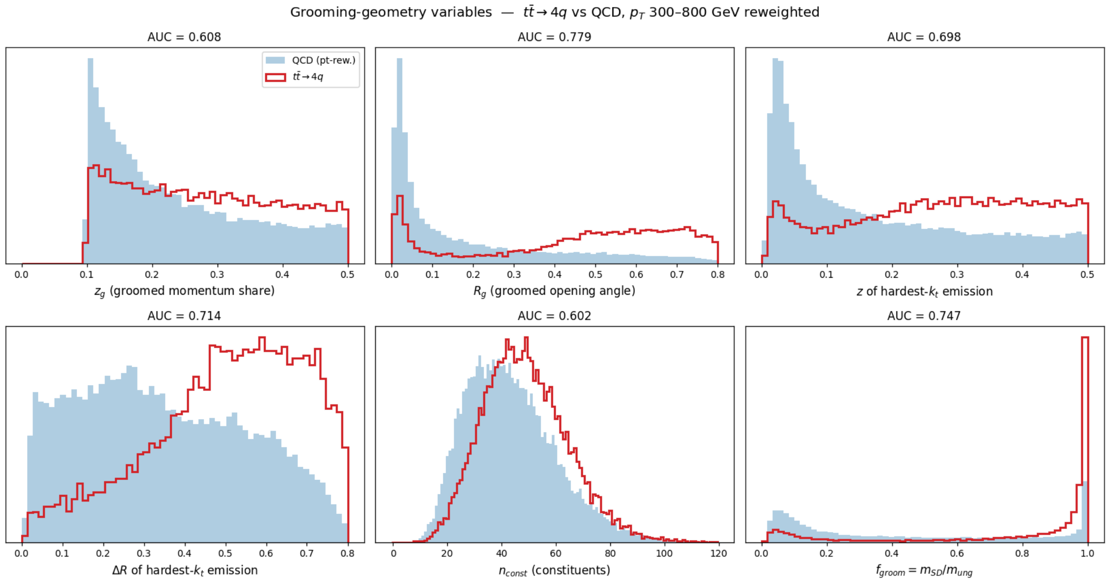
All inputs (3/4) — Lund / counting variables



How many hard emissions, and how hard the sub-leading ones are — the **declustering sequence**, information a single mass number cannot carry. All from the **C/A** primary declustering.

- C/A n_{Lund} (0.602), $n(k_t > 1)$ (0.587): primary-emission counts. Signal has *fewer* despite higher pileup — physical.
- C/A $n(k_t > 5)$ (0.663): counting only *hard* emissions — top/W give 1–2, QCD more.
- C/A $\ln k_t^{(2)}$ (0.688): 2nd-hardest emission — signal bump = the **second decay splitting** (top → W → q$\bar{\text{q}}$).
- C/A $\ln k_t^{(3)}$ (0.613): 3rd emission — weaker.
- C/A n_{drop} (0.769, **best non-mass var**): soft-drop declustering count. Decays pass in **0–1** steps; QCD needs up to ~12.

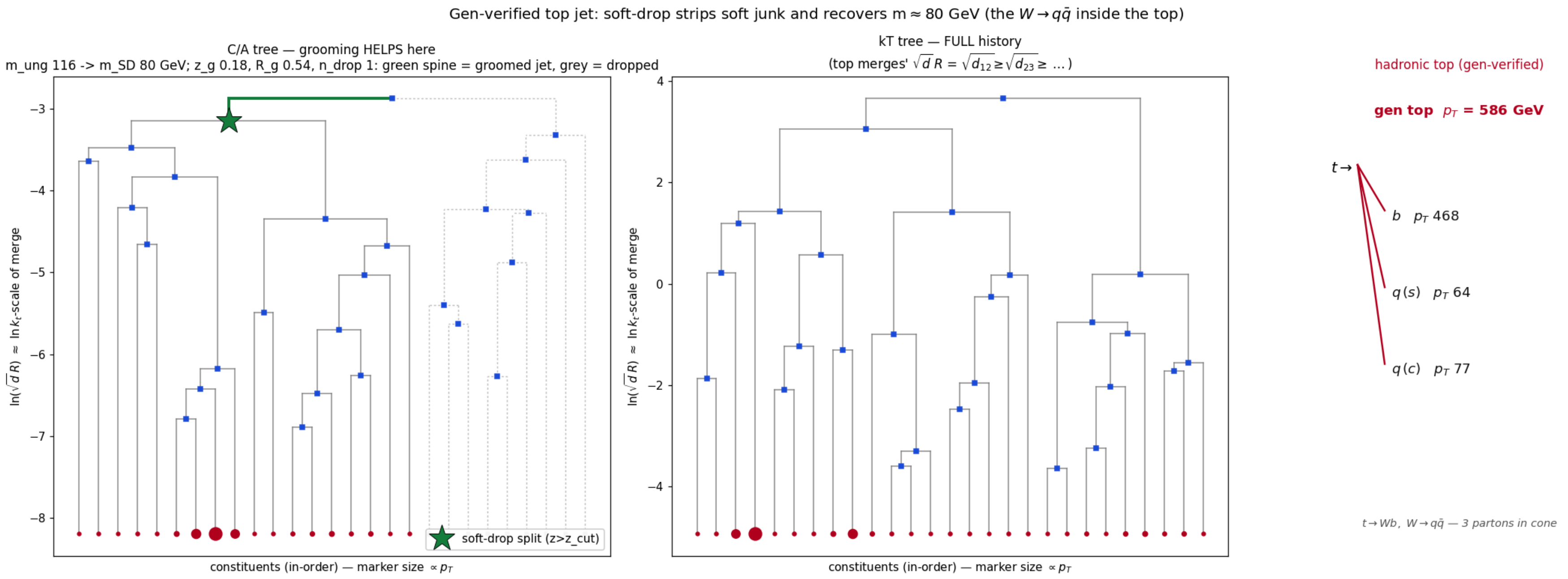
All inputs (4/4) — grooming-geometry variables



Geometry of the split soft drop keeps, plus constituent count and grooming survival.

- **C/A** z_g (0.608): groomed momentum share — near the $z_{cut} = 0.1$ edge; modest alone.
- **C/A** R_g (0.779): groomed opening angle, **strong**. Decay angle fixed and wide; QCD collinear.
- **C/A** z (0.698), ΔR (0.714) **of the hardest-k_t emission**: for a decay this *is* the decay split.
- **anti-kT** n_{const} (0.602): constituent multiplicity — quark/gluon-like, weak alone.
- **C/A+anti-kT** $f_{groom} = m_{SD}/m_{ung}$ (0.747): **grooming survival** — decays keep mass (≈ 1), QCD loses it ($\ll 1$).

What the variables measure: one jet's full merge history (gen-verified)

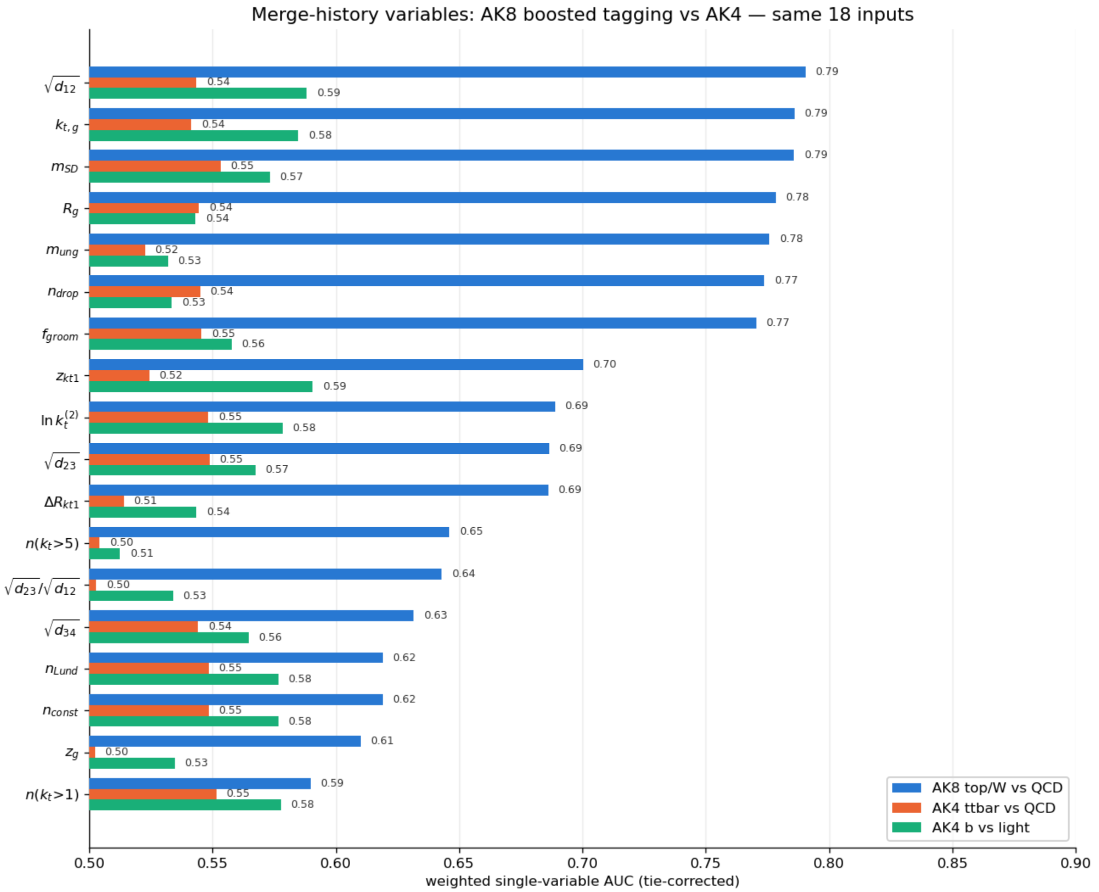


One gen-verified top jet, read three ways — flashjet keeps the *complete* merge tree (`hist_p1,p2,child,d`), nothing thrown away, and every substructure variable is a coordinate on it:

- **Right = gen truth:** `GenPart` → hadronic top ($t \to Wb$, $W \to q\bar{q}$), all three hard partons inside the cone. *What the jet actually is.*
- **Left = C/A tree + grooming: green = surviving groomed jet**, grey = dropped soft prongs, ★ = the passing split — m_{SD}, z_g, R_g **live here.**
- **Middle = kT tree**, splits value-sorted $\sqrt{d_{12}} \geq \sqrt{d_{23}} \geq \ldots$ — source of $\sqrt{d_{12}}$, z_{kt}, the Lund counts.

Punchline: grooming is a *pruned path* through the C/A tree; the ★ it stops on is the real $W \to q\bar{q}$ split, so m_{SD} ($116 \to$**80 GeV**) lands on the W inside the top — confirmed by gen.

AK8 vs AK4: the history is a boosted-decay tool



All 18 merge-history variables, weighted single-variable AUC. **Blue = AK8 top/W-vs-QCD (0.6–0.79)**, orange = AK4 ttbar-vs-QCD, green = AK4 b-vs-light (**both ≈0.5–0.59**) — same variables, strongly discriminating on boosted AK8, near-blind on AK4.

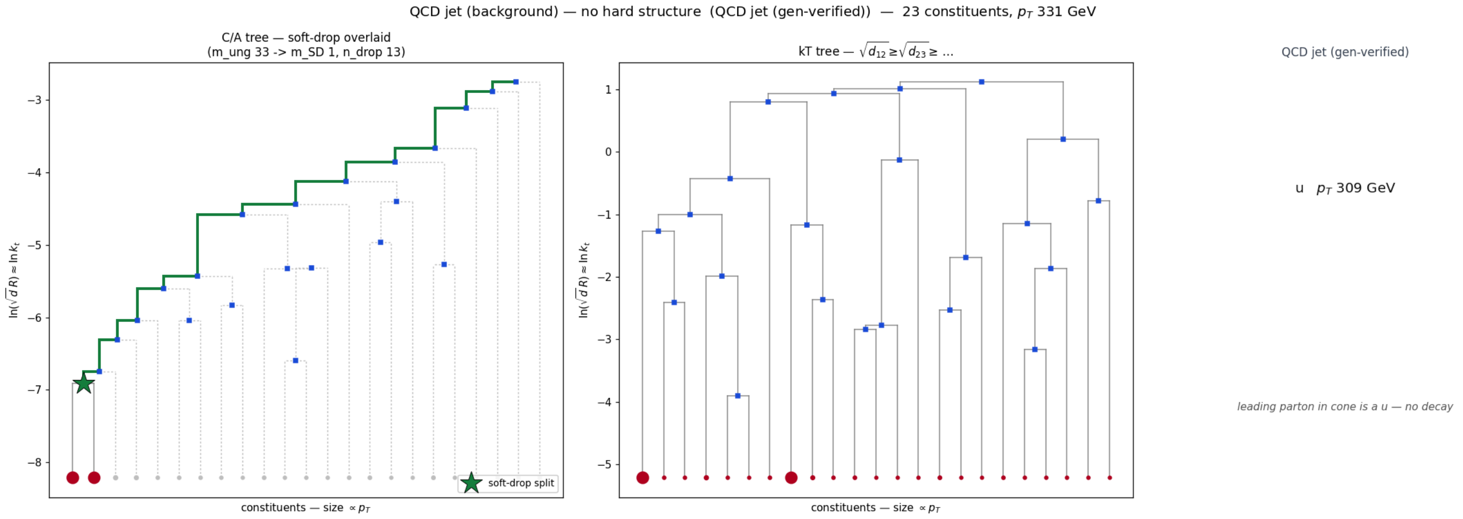
⇒ The history separates boosted 2-/3-prong decays — it does *not* separate AK4 flavour: UParT's impact-parameter / secondary-vertex inputs carry information no clustering tree contains.

AK8 vs AK4 — reading the three studies

study	what's being separated	best variable	AUC	reading
AK8 top/W vs QCD	boosted 2-/3-prong decay vs QCD	$\sqrt{d_{12}}$	**0.79**	history separates **strongly**
AK4 ttbar vs QCD	single quark jet (dijet) vs QCD	m_{SD}	**0.55**	one AK4 jet ≈ one quark — no in-jet decay
AK4 b vs light	flavour (b vs udsg)	z hardest emis.	**0.55**	a **lifetime** question (IP/SV) — invisible to a kinematic tree

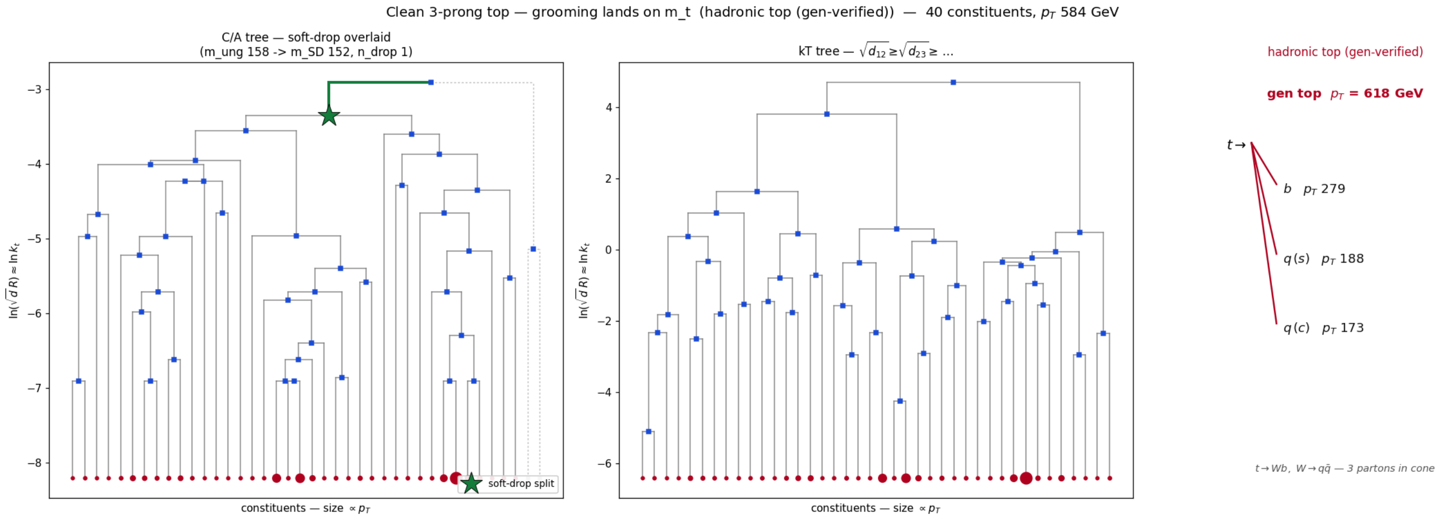
Full AK4 trees + b-vs-light / ttbar-vs-QCD variable studies on the next slides.

Tree gallery 1/4 — QCD jet: a soft staircase, no decay



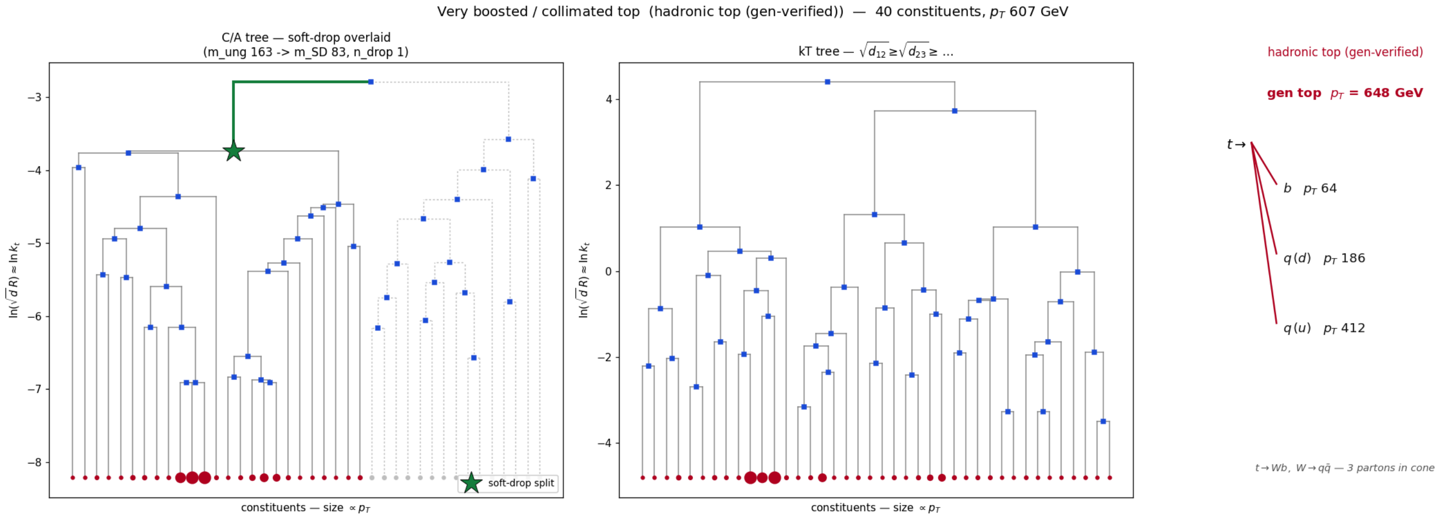
Gen: leading parton is a single light quark, **no resonance**. C/A tree: the green groomed spine is a long **staircase** — soft prong after soft prong stripped, so n_{drop} =**13** and the mass collapses 33 →**1.2 GeV**. There is no wide balanced split to stop on. **This is why n_{drop} is the strongest non-mass discriminant** — QCD has a long spine, a real decay has a short one.

Tree gallery 2/4 — clean top: grooming lands on m_t



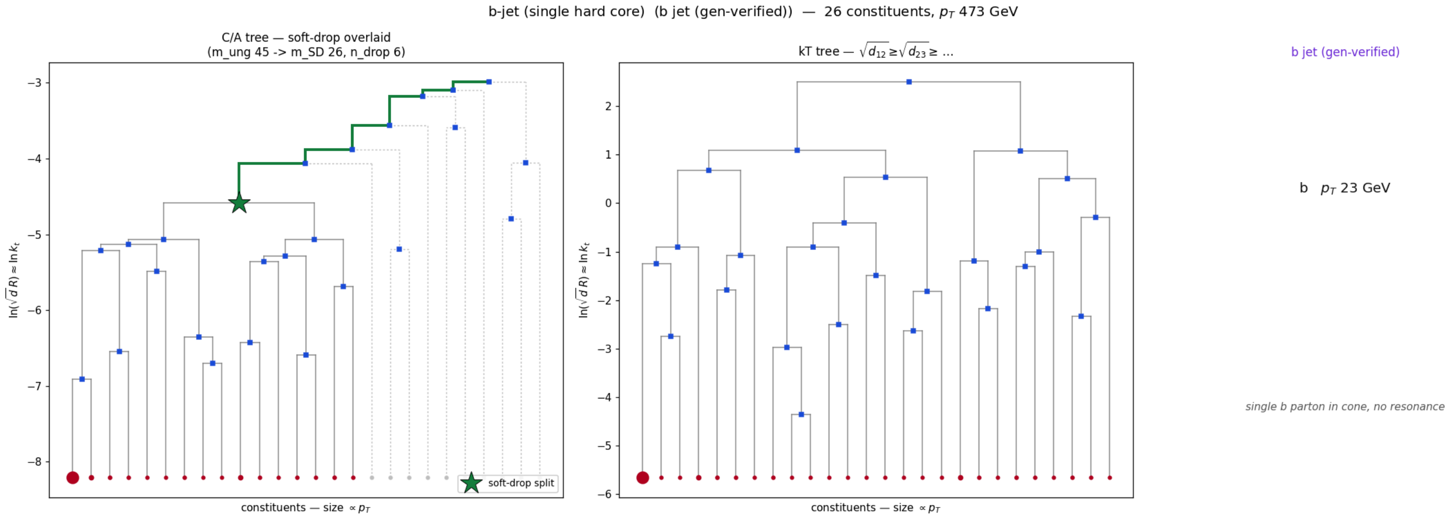
Gen: hadronic top, all of $b+q+q$ in the cone (gen top p_T 618). C/A tree: the ★ sits **near the top of the tree on one wide, balanced split** — grooming barely has to prune, so m_{ung} $158 \rightarrow m_{SD}$ **152 GeV**, n_{drop} $=$**1**. The groomed jet keeps the whole $t \rightarrow bW$ system, so m_{SD} sits on the **top mass**.

Tree gallery 3/4 — boosted top: the same decay, compressed



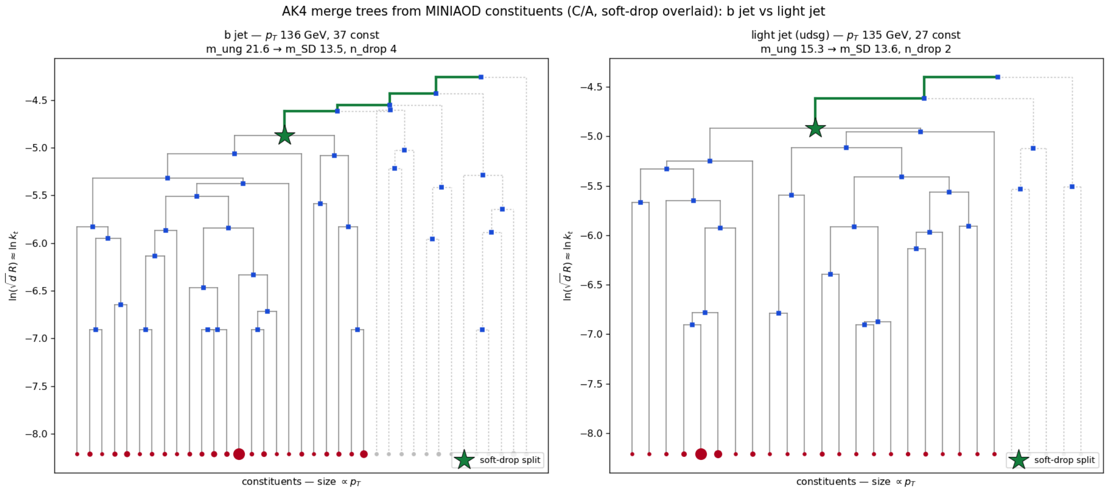
Gen: hadronic top again (gen top p_T 648), but higher-p_T. The decay opening angle shrinks with boost, so the hard split moves *down* into the collinear region and the tree **compresses**: $R_g =$**0.30**, and here grooming resolves the W subsystem, $m_{ung}\,163 \rightarrow m_{SD}$ **83 GeV**. Same physics as slide 2/4 — R_g is what encodes the boost.

Tree gallery 4/4 — b jet: hard core, but looks like QCD



Gen: a single **b** parton in the cone, no resonance. The tree has one hard core and **no balanced two-prong split** — kinematically it is indistinguishable from QCD. Flavour is a *lifetime* question (displaced vertex / impact parameter), which a clustering tree simply cannot see. **This foreshadows the AK4 result: the merge history does not tag flavour.**

AK4 jets — from MINIAOD (constituents NanoAOD doesn't link)



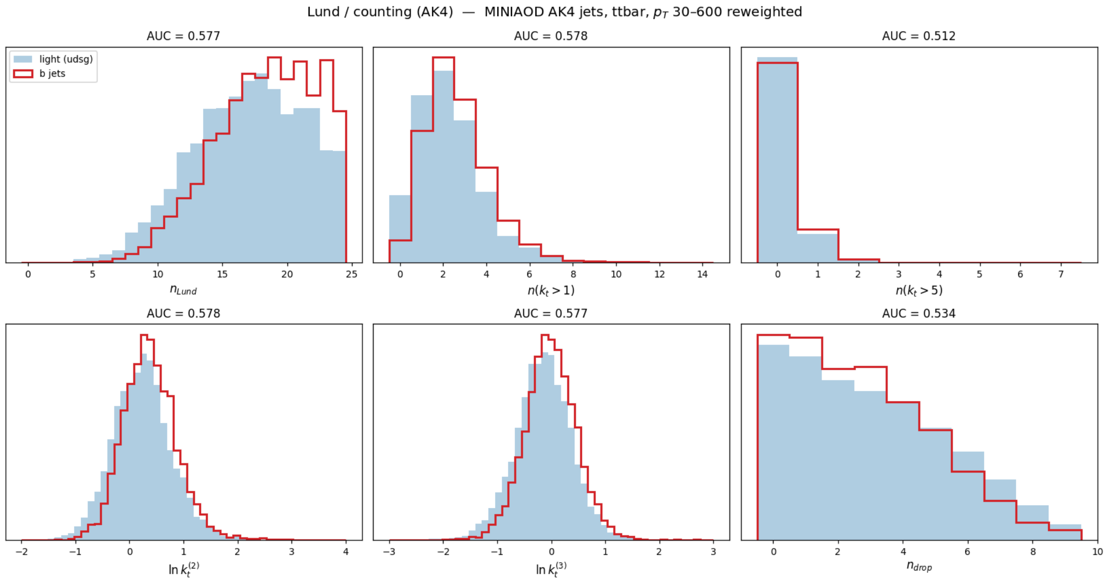
JMENano has AK4 `Jet_*` + `hadronFlavour` but **no PF→AK4 linker** (only `FatJetPFCand` for AK8). So AK4 trees are impossible there. Went to **MINIAOD** (`slimmedJets` carry `packedPFCandidates`) via DAS: 12 000 AK4 jets, **3552 b / 6593 udsg**.

Two-stage pipeline (CMSSW py3.9 and `b_hive` py3.11 are ABI-incompatible):

1. **FWLite** reads `slimmedJets` + PF daughters + `hadronFlavour` → constituent npz
2. **`b_hive`** runs flashjet C/A + kT + soft-drop + Lund → 18 vars + flavour

The b jet and light jet at the *same* p_T have **near-identical** trees.

AK4 result: no flavour separation (b vs light)

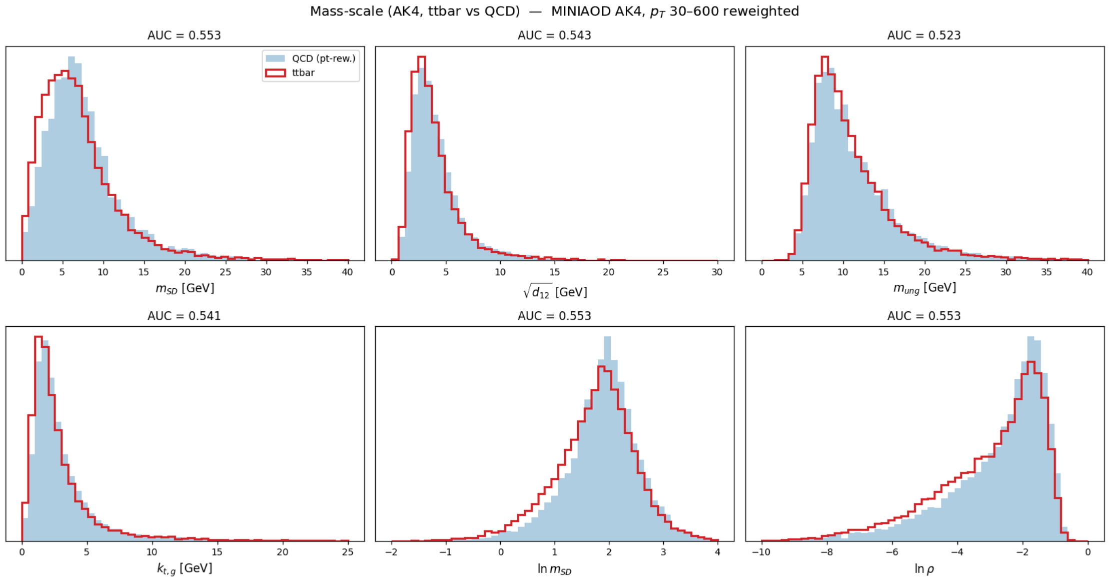


b vs light (udsg), p_T-reweighted — **every variable 0.50–0.59**:

best AK4 variables	AUC
z of hardest emission	0.591
$\sqrt{d_{12}}$	0.588
m_{SD}	0.573
$n(k_t > 5)$, f_{32}	~0.51

b vs light is a **lifetime** question (displaced tracks, SVs) — absent from a kinematic tree. Confirms UParT's IP/SV inputs carry what no tree contains.

AK4 result: ttbar vs QCD (dijet)



ttbar AK4 vs QCD AK4 (MINIAOD, same source) — **every variable 0.50–0.55**:

- best: m_{SD} 0.553, $n(k_t > 1)$ 0.551, $\sqrt{d_{23}}$ 0.549;
- a single AK4 jet in ttbar is mostly **one quark** (a b, or a light quark from $W \to q\bar{q}$) — no multi-prong decay *inside one jet* for the history to see.

The tiny edge is just the b-jet admixture. Complements the b-vs-light null: the history has no AK4 handle at all.

Summary

flashjet reads the full merge history — F1 exclusive-kt jets, F2 soft-drop grooming, F3 Lund — as pure-torch post-reads, validated **jet-by-jet against CMS's FastJet output** on QCD and ttbar (raw-to-raw), with residuals **at the level of NanoAOD float storage**.

Those same histories yield variables that separate the samples: they cleanly split boosted top/W jets from QCD (the declustering *sequence*, not just mass), and the study localizes where they help — **boosted 2-/3-prong decays**, not AK4 flavour.

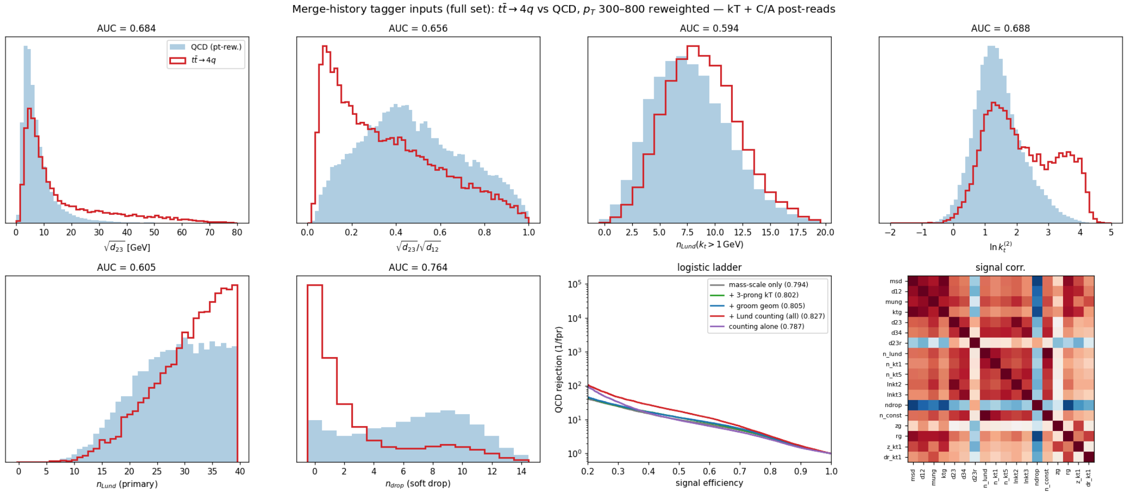
References

1. M. Cacciari, G. P. Salam, G. Soyez, *FastJet User Manual*, Eur. Phys. J. C 72 (2012) 1896 — arXiv:1111.6097. **F1** exclusive jets (`d_cut` / `n_jets`).
2. M. Cacciari, G. P. Salam, G. Soyez, *The anti-k_t jet clustering algorithm*, JHEP 04 (2008) 063 — arXiv:0802.1189. The `hist_d` distance measure, $p = -1/0/+1$ family, jet areas.
3. J. M. Butterworth, A. R. Davison, M. Rubin, G. P. Salam, *Jet substructure as a new Higgs-search channel*, PRL 100 (2008) 242001 — arXiv:0802.2470. **F2** original mass-drop (μ) tagger.
4. A. J. Larkoski, S. Marzani, G. Soyez, J. Thaler, *Soft Drop*, JHEP 05 (2014) 146 — arXiv:1402.2657. **F2** the $z > z_{\text{cut}}(\Delta R/R)^{\beta}$ condition, z_g prediction, β-ordering.
5. M. Dasgupta, A. Fregoso, S. Marzani, G. P. Salam, *Towards an understanding of jet substructure (mMDT)*, JHEP 09 (2013) 029 — arXiv:1307.0007. **F2** the $\beta = 0$ default.
6. F. A. Dreyer, G. P. Salam, G. Soyez, *The Lund Jet Plane*, JHEP 12 (2018) 064 — arXiv:1807.04758. **F3** the $(z, \Delta R, k_t, \ln 1/\Delta R, \ln k_t)$ coordinates.

Broader review: S. Marzani, G. Soyez, M. Spannowsky, *Looking Inside Jets*, arXiv:1901.10342. Papers + a print-ready reader catalogued in `References/Flashjet/papers.md`.

Backup

Multivariable separation: the declustering *sequence* is the payload

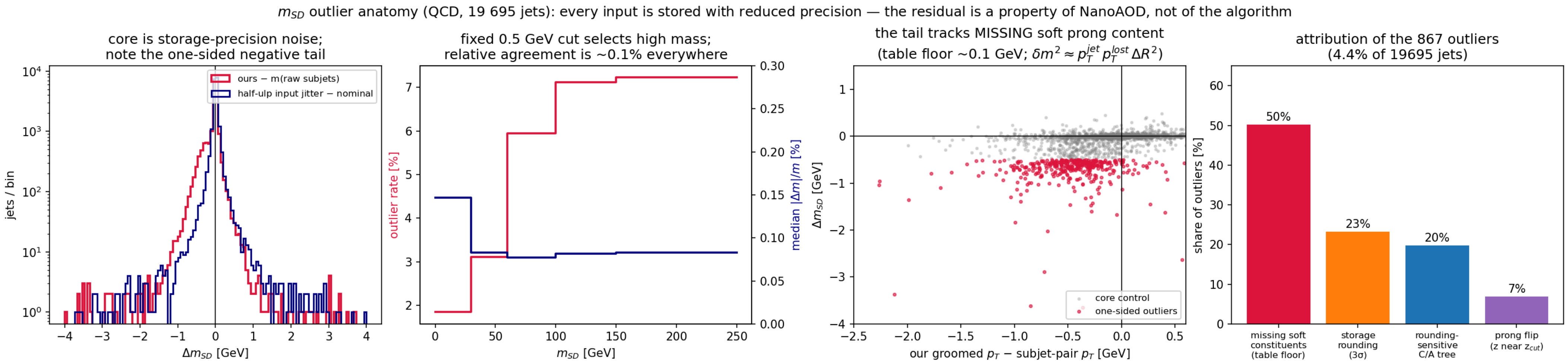


$t\bar{t} \to 4q$ (C3) vs p_T-reweighted QCD, weighted logistic on the **18-variable** history vector:

- mass-scale vars saturate at AUC **0.794** (they're 0.8–0.97 correlated — the *same* 2-prong mass);
- adding the **declustering sequence** → full set **0.827**, ~2× QCD rejection at 30% eff;
- n_{drop} **alone 0.769** — decay jets pass soft drop in 0–1 steps, QCD needs many;
- $\ln k_t^{(2)}$ resolves the **second** decay splitting (top → W).

The point: this variable set is **free** once the history exists — no new clustering, GPU-batchable. Exploratory (no gen-match, linear model). Condor 9128460.

CMS (2b) — anatomy of the residual m_{SD} tail (the 4.4%)



Every input branch is stored with reduced mantissa (`PFCand_pt` ~10 bits ≈ 10^{-3}, `SubJet_pt/mass` ~9 bits, `SubJet_rawFactor` ~5 bits ≈ 2%) while CMS ran FastJet at full precision.

Best-estimate attribution (per-jet tests, 19 695 jets): **~50%** — soft candidates **missing from `FatJetPFCand`** (the table has an effective ~0.1 GeV floor; the tail is one-sided, its groomed-p_T deficit correlates with Δm, and $\delta m^2 \approx p_T^{jet} p_T^{lost} \Delta R^2$ matches);

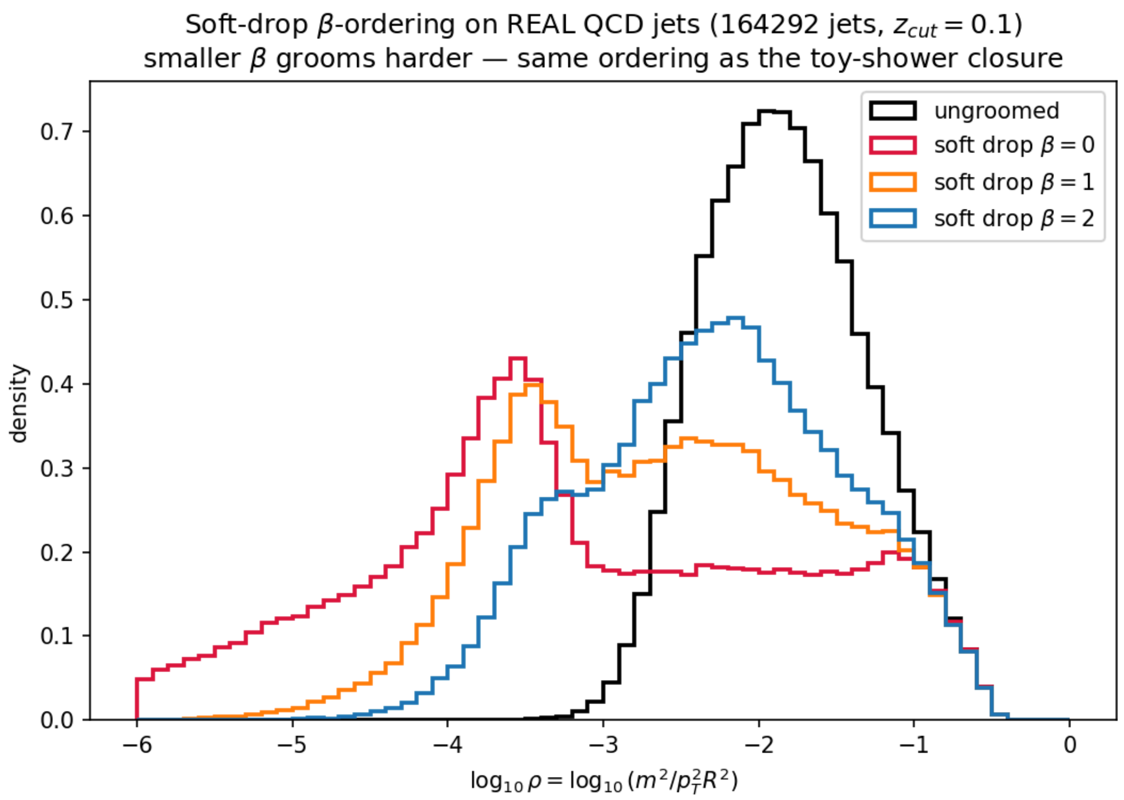
~23% within 3σ of storage rounding; **~20%** rounding-sensitive C/A trees (half-ulp jitter moves them); **~7%** genuine $z \approx z_{\rm cut}$ prong flips (14× enriched vs core). These are **input-level (NanoAOD) effects, not an algorithm error** — relative agreement is ~0.1% everywhere; we don't claim the split is exact. Scripts: `outliers.py` (HTCondor 9098953).

Soft-drop β-family on real QCD jets (F2) — C1

The toy-shower closure, repeated on **164 292 real CMS jets**: re-groom the same C/A trees at $\beta = 0, 1, 2$ ($z_{\mathrm{cut}} = 0.1$) and plot $\rho = m^2/(p_T^2 R^2)$.

- grooming pushes mass **down**; **smaller β grooms harder** — the exact ordering of Soft Drop [4] Figs. 3–4;
- $\beta = 0$ (mMDT) develops the characteristic flat low-ρ tail.

Grooming re-run 3× on the *same* merge histories (a pure post-read: no re-clustering — the point of the history design).



Correction: AUC tie handling

The AUC routine integrated the ROC **without handling ties**. For discrete counts (>80% of jets share one value) that splits tied jets arbitrarily and **manufactures separation**. Caught because AK4 $n(k_t > 5)$ scored **0.773** while b and light had *identical means in every p_T slice* — impossible.

Fixed by building the ROC on **unique values** (proper Mann–Whitney tie handling). Continuous variables unaffected; corrected AK8 counts:

variable	was	corrected
$n(k_t > 1)$	0.593	**0.587**
$n(k_t > 5)$	0.691	**0.663**
n_{drop}	0.765	**0.769**
f_{match}	0.516	**0.648**

All mass/geometry variables (m_{SD} 0.782, $\sqrt{d_{12}}$ 0.792, R_g 0.779) **unchanged** — AK8 conclusions stand.

Reproducibility & full-statistics numbers

All CMS plots regenerated **raw-to-raw on the C/A tree** at full statistics (`make_cms_plots.py`, HTCondor cluster 9087059, 60 257 jets):

observable	comparison	result
jet p_T (QCD)	our anti-kt vs `FatJet_pt×(1−rawFactor)`	median **1.000000**, σ 2.5×10^{-4}
soft-drop mass (QCD)	our C/A-tree vs m(raw sub_1+sub_2)	median **−0.004 GeV**, 95.7% <0.5 GeV
z_g (QCD)	our vs raw subjet z	$\lvert\Delta\rvert$ = $\mathbf{7\times10^{-5}}$
jet p_T (ttbar 2L2Nu)	12 561 leading jets, HTCondor 9098883	median **1.000002**, σ 2.2×10^{-4}
soft-drop mass (ttbar)	our C/A-tree vs m(raw sub_1+sub_2)	median **−0.041 GeV**, 94.2% <0.5 GeV
full-event (QCD)	all PFCands → anti-kt, ΔR-match	7 701 jets, pt **1.0000**, ΔR med **0.0019**
R_g (3 samples)	our groomed `dR` vs ΔR(sub_1,sub_2)	median Δ ≤ 2×10^{-4}; 99.2/95.9/87.0% <0.01

Residuals are consistent with NanoAOD float storage throughout (see CMS(2b) for the breakdown). Jets: anti-kt $R = 0.8$; grooming/Lund: big-R **C/A** reclustering (as FastJet).

Function reference (added)

```
exclusive_jets_from_history(hist_p1, hist_p2, hist_child, hist_d, mask,
                            n_jets=None, d_cut=None)          # F1
groom_from_history(hist_p1, hist_p2, hist_child, hist_d, mask, p4, R,
                   z_cut=0.1, beta=0.0, mu=None)              # F2 (soft-drop / mMDT / mass-drop)
lund_coordinates_from_history(hist_p1, hist_p2, hist_child, hist_d, mask, p4)  # F3 -> (B,J,S,6)

# helpers
_resolve_roots(...)      # pointer-jump roots of an arbitrary sub-forest
_jet_roots(...)          # per-jet root pseudojet id, beam-merge order
_pseudojet_p4(...)       # E-scheme 4-momentum of every pseudojet, keyed by id
_dense_number_roots(...) # compact per-particle jet index
```

API surface: `ClusterOutput.exclusive_jets`, `.groomed_jets`, `.mass_drop`, `.lund_coordinates`, plus a new `.mask` field to map slots→ids.

Reproducing everything

```
# on lxplus, env b_hive (torch 2.5.1)
cd /eos/home-c/cgupta/flashjet/plots/2026-07-13-substructure
micromamba run -n b_hive python make_plots.py        # F1/F2/F3 toy justification + parity
micromamba run -n b_hive python make_paper_plots.py  # anti-kt/SoftDrop/Lund paper figures
micromamba run -n b_hive python make_cms_plots.py    # real CMS UL18 QCD (C1)
micromamba run -n b_hive python make_ttbar_plots.py  # ttbar (C2)
micromamba run -n b_hive python make_compare_plots.py # C1/C2/C3 comparisons
# AK4/AK8 from MINIAOD: cmsenv in CMSSW_14_1_0_pre4 + voms-proxy-init, then
#   python3 dump_ak4_constituents.py / dump_ak8_constituents.py  (FWLite)
#   micromamba run -n b_hive python ak4_substructure.py / ak4_vs_ak8.py

# tests
cd /eos/home-c/cgupta/flashjet/FlastJetDemo
micromamba run -n b_hive python -m pytest -q          # 85 passed, 13 skipped
```

Fixed seed `20260713` throughout. Papers catalogued in `References/Flashjet/papers.md`.